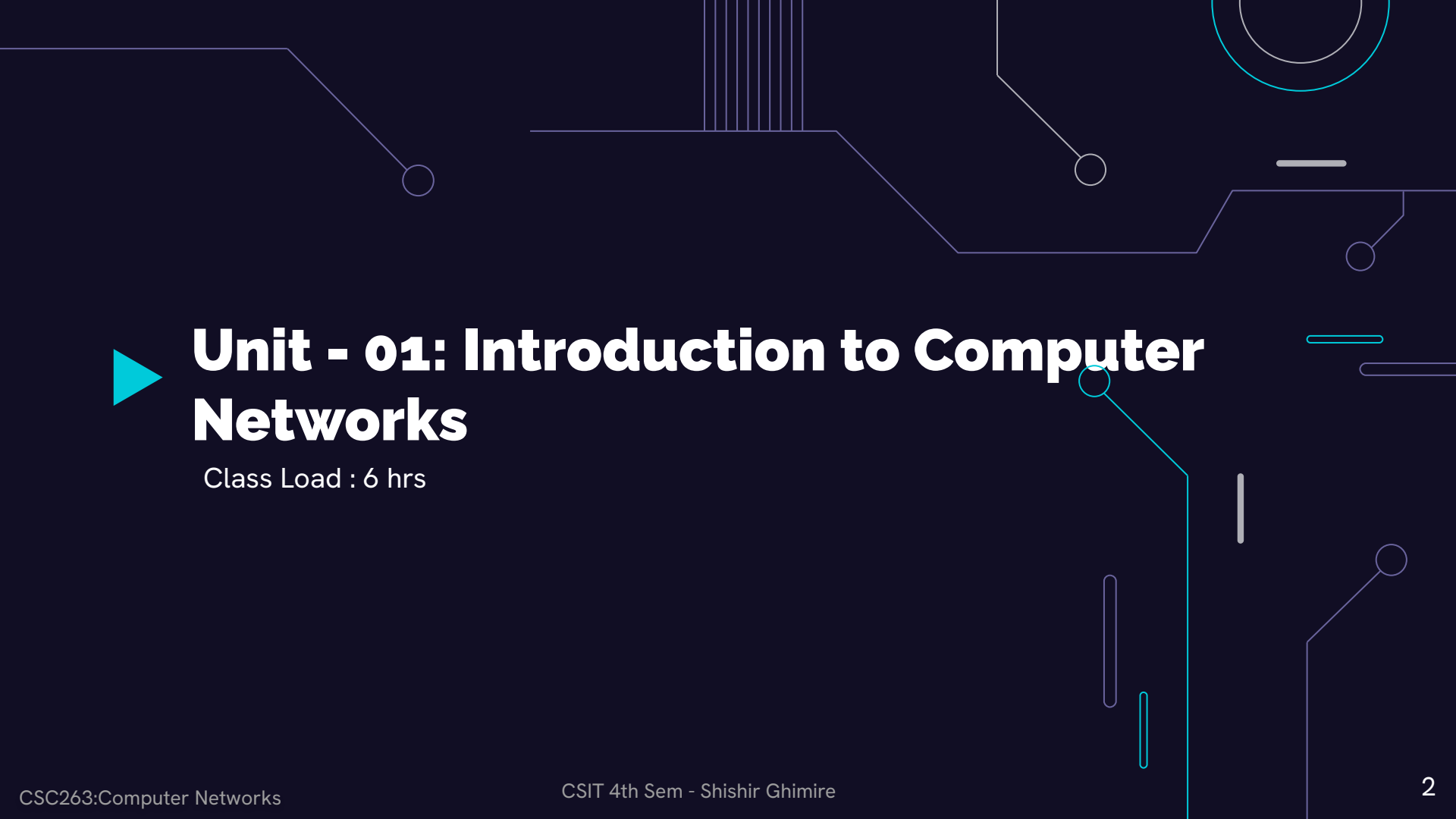


Computer Networks

► **Course Code: CSC263**
Year/ Semester: II/IV

Compiled by Shishir Ghimire

Credit Hours: 3hrs

The background features a dark blue field with intricate white and light blue line art. These lines form a complex network of paths, some ending in small circles, resembling a circuit board or a data network map. There are also some rectangular and circular shapes scattered throughout.

▶ Unit - 01: Introduction to Computer Networks

Class Load : 6 hrs

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Unit 1: Introduction to Computer Network (6Hrs.)

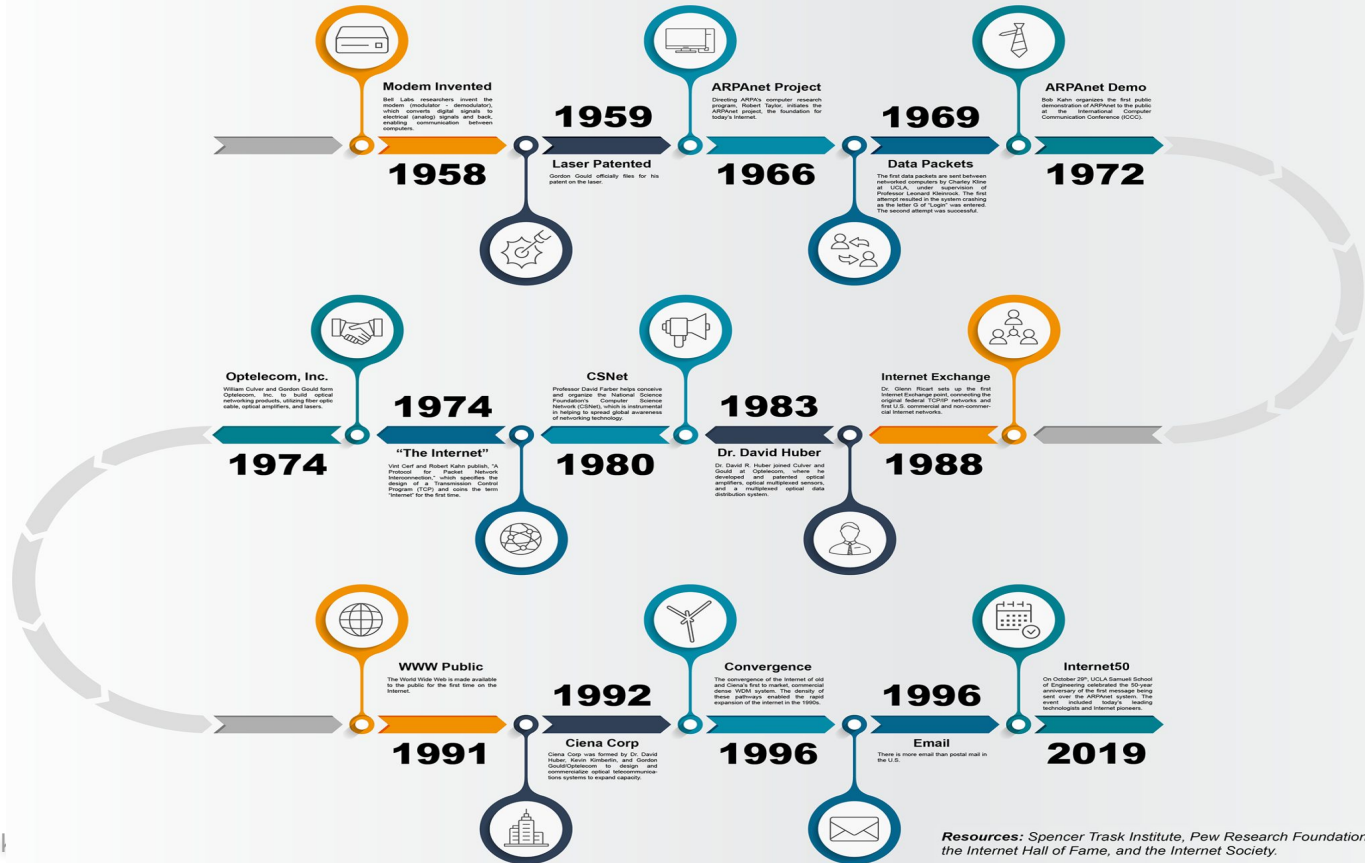
- 1.1. Definitions, Uses, Benefits
- 1.2. Overview of Network Topologies (Star, Tree, Bus,...)
- 1.3. Overview of Network Types (PAN, LAN, CAN, MAN,...)
- 1.4. Networking Types (Client/Server, P2P)
- 1.5. Overview of Protocols and Standards
- 1.6. OSI Reference Model
- 1.7. TCP/IP Models and its comparison with OSI.
- 1.8. Connection and Connection-Oriented Network Services
- 1.9. Internet, ISPs, Backbone Network Overview

► Introduction Session:



Introduction Session:

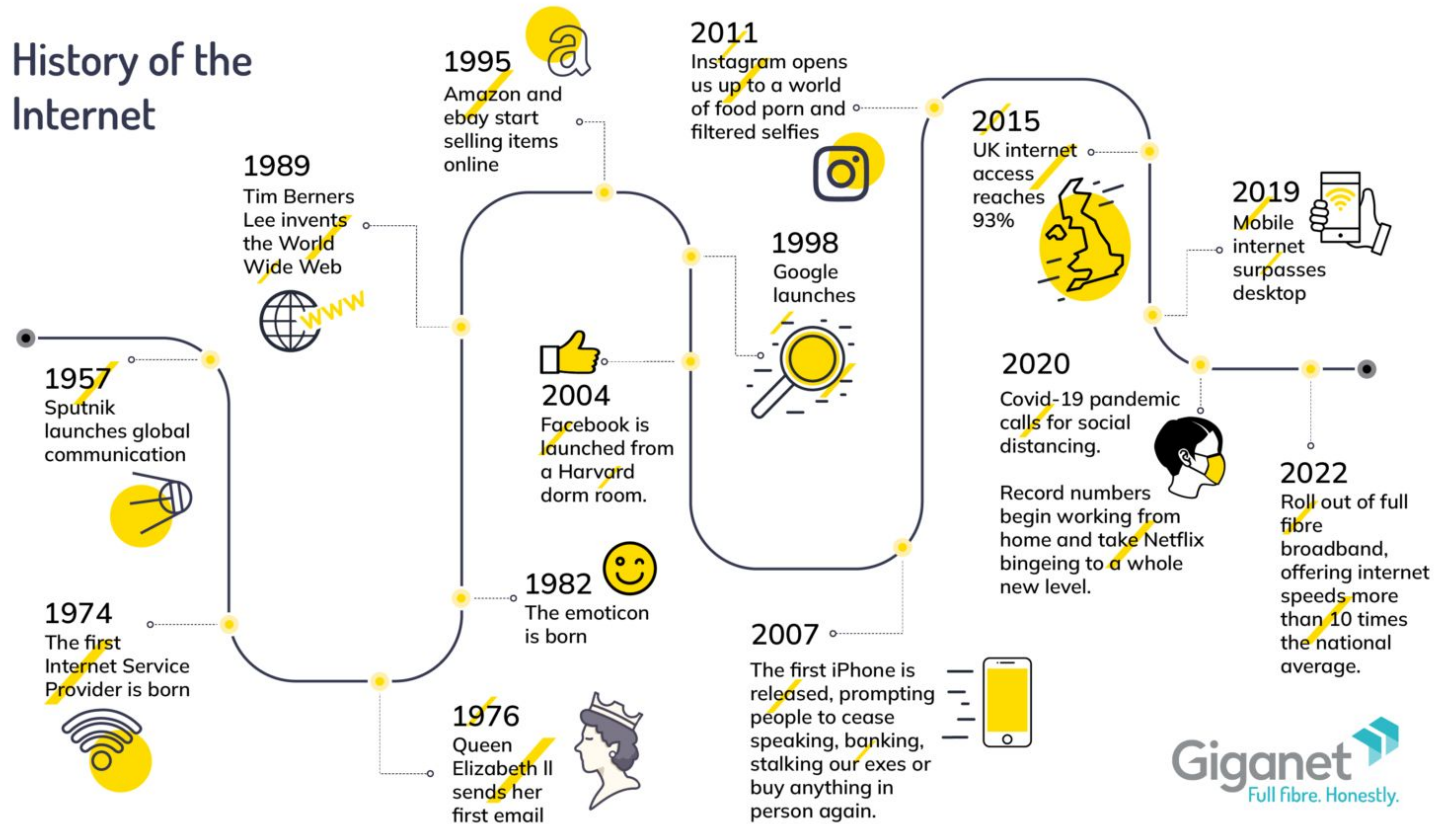
INTERNET HISTORY Timeline



Resources: Spencer Trask Institute, Pew Research Foundation, the Internet Hall of Fame, and the Internet Society.

Introduction Session:

History of the Internet

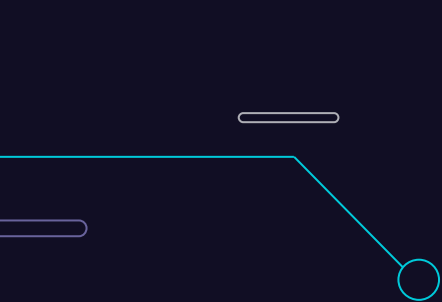


Giganet
Full fibre. Honestly.

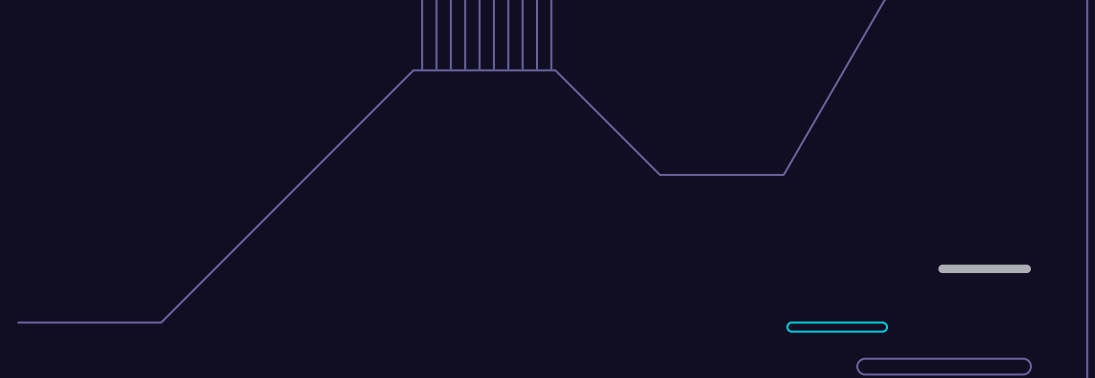
► Quotes on CN:

“All one needs is a computer, a network connection, and a bright spark of initiative and creativity to join the economy.” **-Don Tapscott**

“When you're young, you look at television and think, there's a conspiracy. The networks have conspired to dumb us down. But when you get a little older, you realize that's not true. The networks are in business to give people exactly what they want.” **-Steve Jobs**



Definitions, ► Uses, Benefits



1.1

► Definition:

- **Network:** set of devices or **nodes** connected by communication link.
- A computer network is a collection of computers or devices which are connected to each other by communication link or medium for the **purpose of sharing** their resources.
- Any device which can share or receive the data is called a **Node**.
- Through which the information or data propagate is known as **channels**.
 - It can be guided or unguided.



► Uses:

- **Communication:** Facilitate communication via email, video conferencing, instant messaging, etc.
- **Resource Sharing:** Enable multiple users to share a single hardware device like a printer or scanner.
- **File Sharing:** Enable file sharing across the network.
- **Sharing Information:** Allow for the sharing of software or operating programs on remote systems.
- **Flexible Access:** Make information easier to access and maintain among network users.

► Applications :

1. Business Application:

- Computer Network is widely used in business application such as bank, travel etc.
- Fundamental aim is to share resources i.e software, hardware etc.

2. Home Application:

- Access remote information, person to person communication([instant messaging](#)), interactive entertainment, E-Commerce(B2B, B2C, C2C, G2C, P2P) etc.

3. Mobile Users:

- Mobile Computers such as notebook computers and [mobile phones](#) are one of the fastest growing applications.

4. Social Issues:

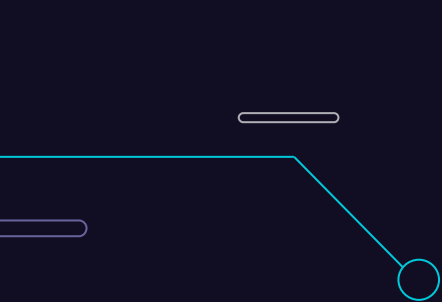
- Computer Network plays an important role in social issues such as [newsgroup](#) or [bulletin](#).

► Benefits:

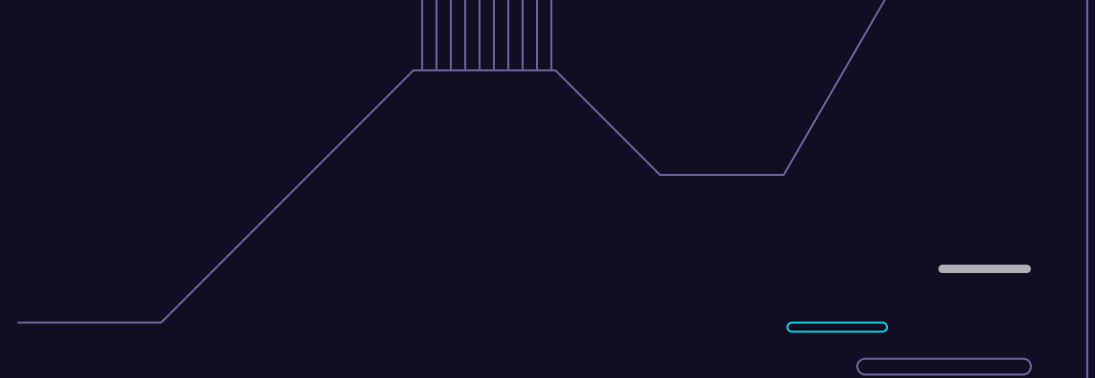
- **Convenient Resource Sharing:** Networked computers can share hardware (like printers) and software, reducing **redundancy and costs**.
- **Easy File Sharing:** Files can be easily transferred between computers, enhancing **collaboration and productivity**.
- **Flexibility:** Networks can adapt to changing needs, allowing for the **addition of new devices and services** without major disruptions.
- **Inexpensive Operation:** Operating costs are lower due to **shared resources and centralized management**, which reduces the need for individual setups.
- **Increased Storage Capacity:** Networks can **pool storage resources**, allowing users to access larger and more diverse data repositories.
- **Cost Efficiency:** Shared resources and services lead to better utilization of **existing assets**, reducing the overall cost per user.
- **Enhanced Communication and Information Availability:** Networks enable **quick and reliable** communication and access to information, facilitating timely decision-making and coordination.

► Disadvantages:

- **Security Risks:** Networks are vulnerable to **unauthorized access**, hacking, and **data breaches**, which can compromise sensitive information.
- **Maintenance Costs:** Regular maintenance and upgrades of network **infrastructure** can be costly and require specialized knowledge.
- **Network Failures:** If the network fails or experiences **downtime**, it can **disrupt** business operations and productivity.
- **Complexity:** Setting up and managing a network can be complex and require **skilled personnel** to ensure proper configuration and troubleshooting.
- **Virus and Malware Spread:** Viruses and malware can **spread quickly** across interconnected systems, affecting multiple devices simultaneously.
- **Dependence on Network:** Users become heavily reliant on the **network**, and any issues can lead to significant inconvenience and **work stoppages**.
- **Privacy Concerns:** Sharing resources and data can raise privacy concerns, as personal or sensitive information may be **accessed by unauthorized users**.



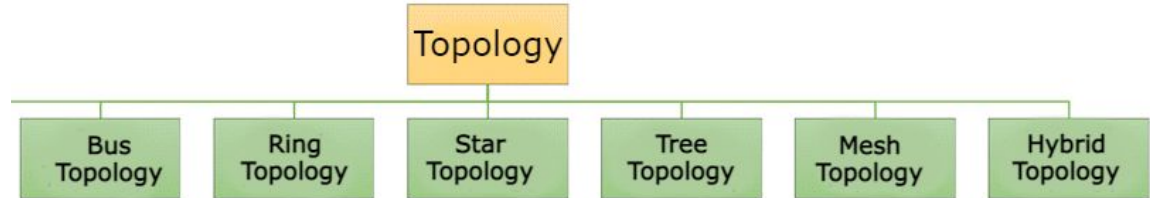
Overview of ► Network Topologies



1.2

► Network Topologies:

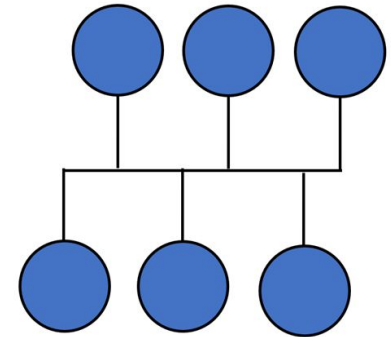
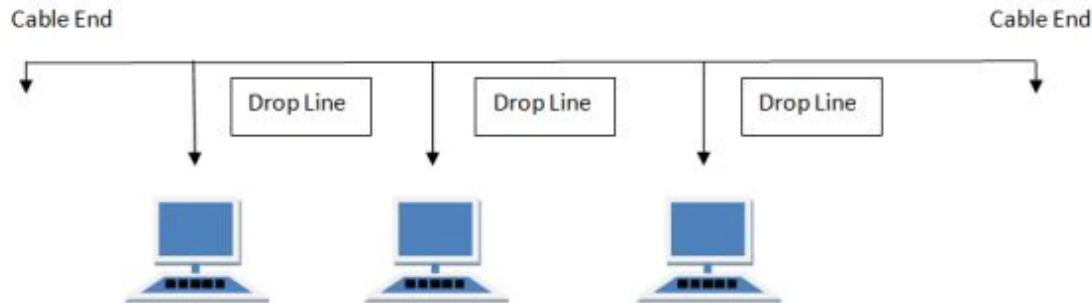
- **Topology:** schematic description of a network arrangement, connecting various nodes (sender and receiver) through lines of connection.
 - **Physical** – actual layout of the computer cables and other network devices.
 - **Logical** – the way in which the network appears to the devices that use it.
- Some Common Topologies:
 - Bus
 - Star
 - Ring
 - Mesh
 - Hybrid
 - Tree
 - Wireless.



► Bus Topology:

Bus topology is a network type in which every computer and network device is connected to single cable known as a **backbone cable**.

- Each node is either connected to the backbone cable by **drop cable or directly** connected to the backbone cable.
- When a node wants to send a message over the network, it **puts a message over the network**. All the stations available in the network will receive the message whether it has been **addressed or not**.



► Bus Topology:

Features of Bus Topology:

- It transmits data only in one direction.
- Every device is connected to a single cable.
- The most common **access method** of the bus topologies is **CSMA (Carrier Sense Multiple Access)** -to solve the issue of **multiple hosts sending data** at same time.

Advantages:

- **Cost-Effective:** Cheap and easy to implement
- Cable required is least compared to other network topology.
- Does not used any specialized network equipment.
- One node failures can't affect the entire system.
- Easy to expand joining two cables together.

► Bus Topology:

Disadvantages:

- Network disruption when computers are added or removed.
- A break in the cable causes system or whole system failures.
- Difficult to troubleshoot.
- It is slower than the ring topology.

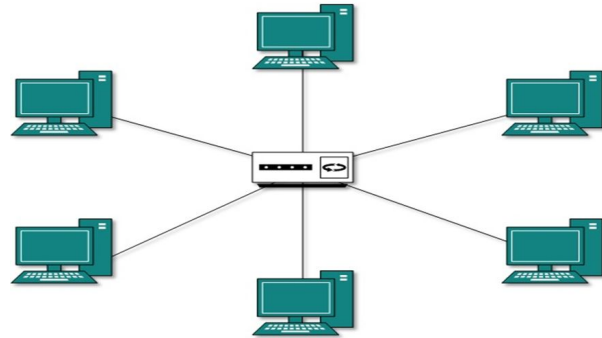
Applications:

1. Local Area Networks (LANs)
2. Industrial Control Systems
3. Building Automation Systems

► Star Topology:

Star topology is an arrangement of the network in which every node is connected to the **central hub**, switch or a central computer.

- The central computer is known as a **server**, and the peripheral devices attached to the server are known as **clients**.
- **Most Widely used Topology.**
- Hub is the single point of failure.



► Star Topology:

Features of Star Topology:

- Every node has its own dedicated connection to the hub.
- Hub acts as a repeater for data flow.
- Can be used with twisted pair, Optical Fibre or coaxial cable.

Advantages:

- Fast performance with few nodes and low network traffic.
- Hub can be upgraded easily.
- Easy to troubleshoot and isolate problems.
- Easy to setup and modify.
- Only that node is affected which has failed, rest of the nodes can work smoothly.

► Star Topology:

Disadvantages:

- Cost of installation is high as it requires more cable.
- Expensive to use.
- If the hub fails then the whole network is stopped because all the nodes depend on the hub.
- Performance is based on the hub that is it depends on its capacity.

Applications:

- Local Area Networks (LANs) in an office where all computers are connected to a central hub.

► Ring Topology:

It is called ring topology because it forms a ring as each computer is connected to another computer, with the **last one connected to the first**.

Exactly two neighbors for each device.

- Ring topology is like a bus topology, but with connected ends.
- The node that receives the message from the previous computer will **retransmit** to the next node.
- The data flows in one direction, i.e., it is **unidirectional**.



► Ring Topology:

Features of Ring Topology:

- A number of **repeaters** are used for Ring topology with **large number of nodes**, because if someone wants to send some data to the **last node** in the ring topology with 100 nodes, then the data will have to pass through 99 nodes to reach the 100th node. Hence to prevent data loss repeaters are used in the network.
- The transmission is **unidirectional**, but it can be made bidirectional by having 2 connections between each Network Node, it is called **Dual Ring Topology**.
- The most common **access method** of the ring topology is **token passing**.

Token passing: It is a network access method in which token is passed from one node to another node.

Token: It is a frame that circulates around the network.

► Ring Topology:

Advantages:

- Transmitting network is **not affected** by high traffic or by adding more nodes, as only the nodes having **tokens** can transmit data.
- Cheap to install and expand.

Disadvantages:

- Troubleshooting is difficult in ring topology.
- Adding or deleting the computers disturbs the network activity.
- Failure of one computer disturbs the whole network

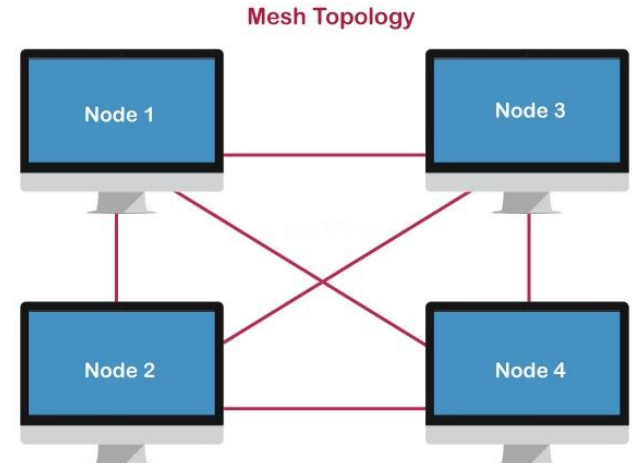
Applications:

1. Local Area Networks (LANs) in an office where all computers are connected in a ring structure.

► Mesh Topology:

Mesh Topology is an arrangement of the network in which computers are interconnected with each other through various **redundant connections**.

- There are **multiple paths** from one computer to another computer.
- It does not contain the **switch, hub or any central computer** which acts as a central point of communication.
- The **Internet** is an example of the mesh topology.
- Mesh topology is mainly used for **WAN implementations** where communication failures are a critical concern.



► Mesh Topology:

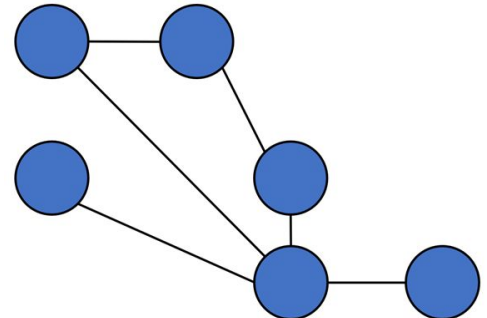
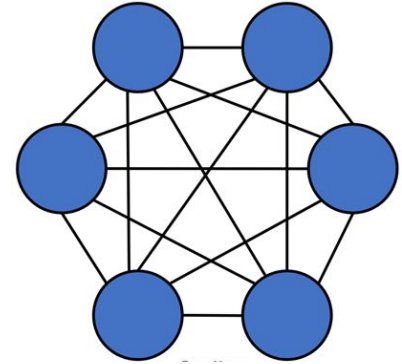
Features of Mesh Topology:

- Mesh topology can be formed by using the formula:

$$\text{Number of cables} = (n*(n-1))/2;$$

Where n is the number of nodes that represents the network.

- Mesh topology is divided into two categories:
 - a. **Full Mesh Topology:** In a full mesh topology, each computer is connected to all the computers available in the network.
 - b. **Partial Mesh Topology:** In a partial mesh topology, not all but certain computers are connected to those computers with which they communicate frequently.



► Mesh Topology:

Advantages:

- **Reliable:** The mesh topology networks are very reliable as if any link breakdown will not affect the communication between connected computers.
- **Fast Communication:** Communication is very fast between the nodes.
- **Easier Reconfiguration:** Adding new devices would not disrupt the communication between other devices.

Disadvantages:

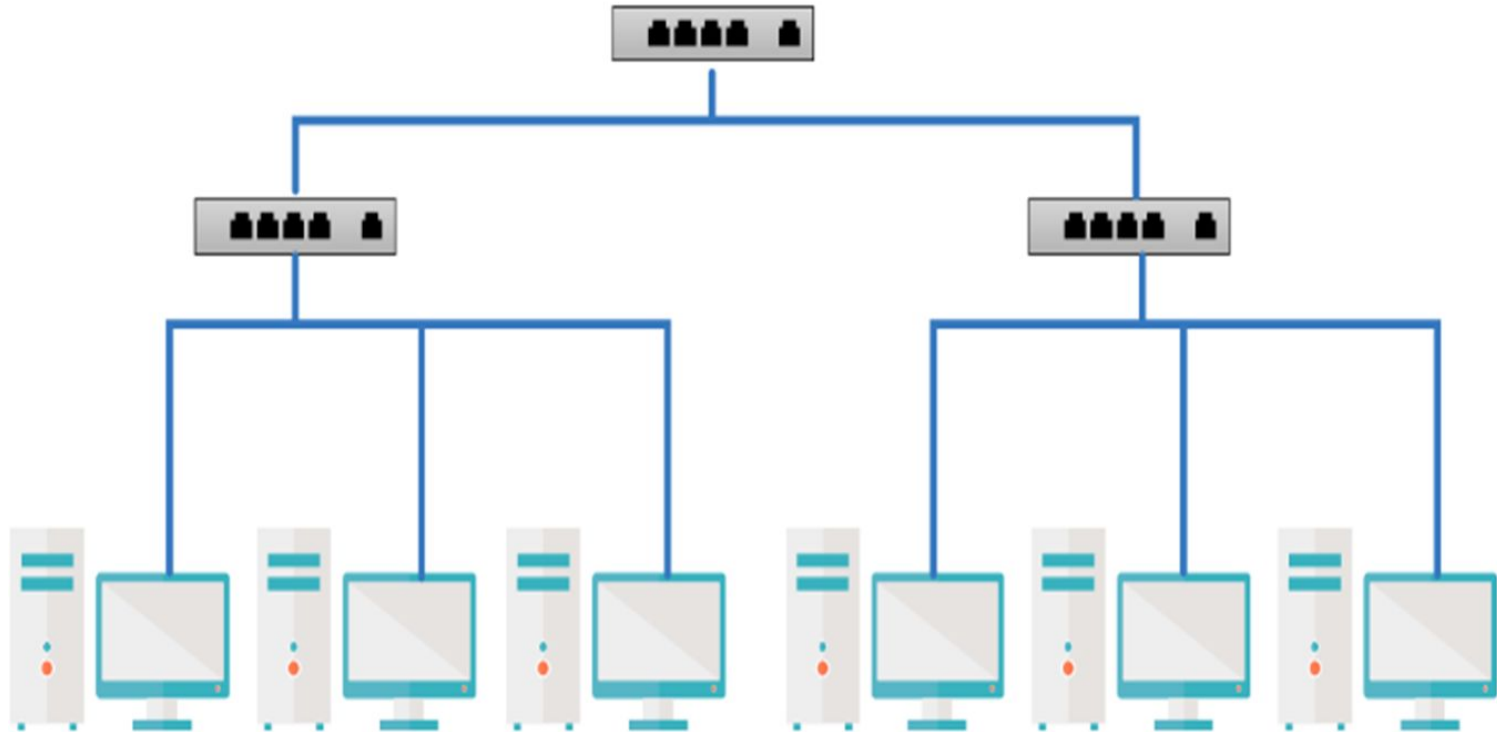
- **Cost:** A mesh topology contains a large number of connected devices such as a router and more transmission media than other topologies.
- **Management:** Mesh topology networks are very large and very difficult to maintain and manage. If the network is not monitored carefully, then the communication link failure goes undetected.
- **Efficiency:** In this topology, redundant connections are high that reduces the efficiency of the network.

► Tree Topology:

A tree topology is a type of structure in which all the computers are connected with each other in **hierarchical fashion**.

- In a tree topology, there is one **central node** (the “trunk”), and each node is connected to the central node through a **single path**.
- This topology **integrates various star topologies** together in a single bus, so it is known as a **Star Bus topology**.
- Tree topologies are often used to create **large** networks.
- There is only one path exists between two nodes for the data transmission. Thus, it forms a **parent-child hierarchy**.

► Tree Topology:



► Tree Topology:

Advantages:

- **Support for broadband transmission:** Tree topology is mainly used to provide broadband transmission, i.e., signals are sent over long distances without being attenuated.
- **Easily expandable:** We can add the new device to the existing network. Therefore, we can say that tree topology is easily expandable.
- **Easily manageable:** In tree topology, the whole network is divided into segments known as star networks which can be easily managed and maintained.
- **Limited failure:** The breakdown in one station does not affect the entire network.
- **Point-to-point wiring:** It has point-to-point wiring for individual segments.

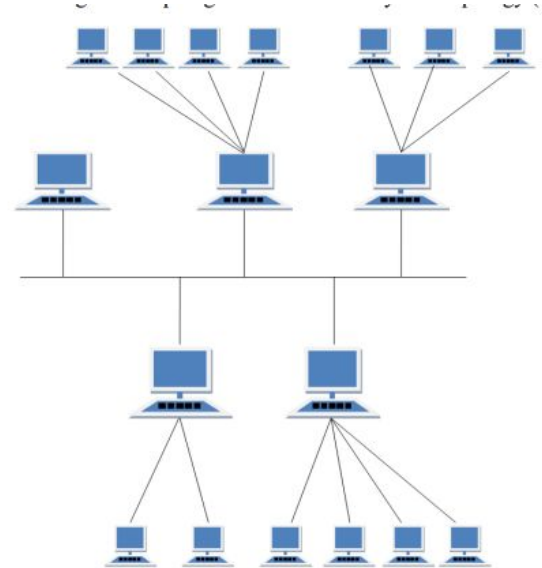
Disadvantages:

- **Difficult troubleshooting:** If any fault occurs in the node, then it becomes difficult to troubleshoot the problem.
- **High cost:** Devices required for broadband transmission are very costly.
- **Failure:** A tree topology mainly relies on main bus cable and failure in main bus cable will damage the overall network.
- **Reconfiguration difficult:** If new devices are added, then it becomes difficult to reconfigure.

► Hybrid Topology:

Hybrid topology is a type of network topology in which two or more **different topologies** are integrated or combined to lay out a network.

- When two or more different topologies are combined together is termed as Hybrid topology and if **similar topologies** are connected with each other will not result in Hybrid topology.
- Inherits the advantages and disadvantages of the topologies included.
- **For example**, if there exist a ring topology in one branch of NICA bank and bus topology in another branch of NICA bank, connecting these two topologies will result in Hybrid topology.



► Hybrid Topology:

Advantages:

- Reliable as Error detecting and troubleshooting is easy.
- Effective.
- Scalable as size can be increased easily.
- Flexible

Disadvantages:

- The design of hybrid topology is complex
- It is one of the costliest processes

► Class Discussion / Question:

If there are 'N' devices (nodes) in a network, what is the number of cable links required for a fully connected mesh and a star topology respectively.

(A) $N*(N-1)/2$, $N-1$

(B) N , $N-1$

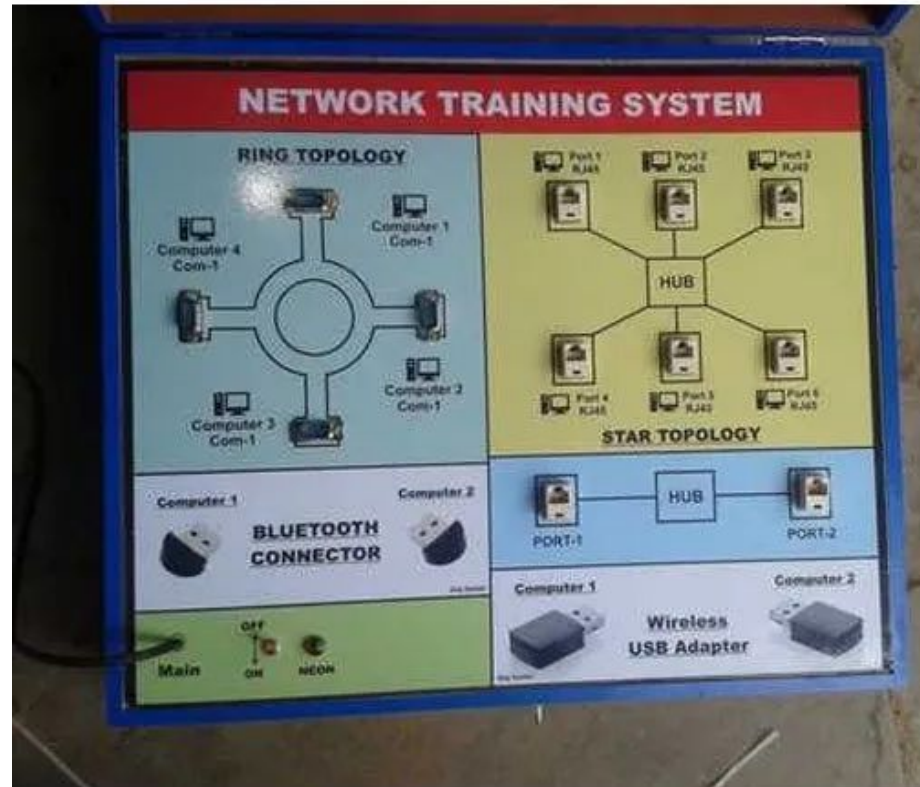
(C) $N-1$, N

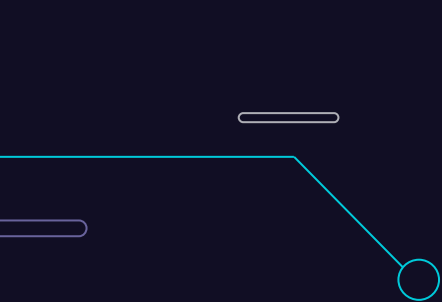
(D) $N-1$, $N*(N-1)/2$

► Summary:

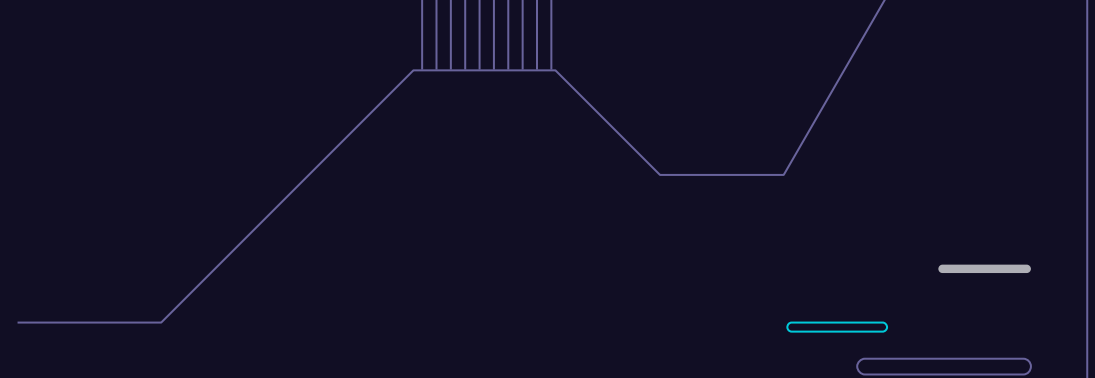
Topology	What it is?
Bus	Uses a single cable which connects all the included nodes
Ring	Every device has exactly two neighboring devices for communication purpose
Star	All the computers connect with the help of a hub.
Mesh	The mesh topology has a unique network design in which each computer on the network connects to every other.
Tree	Tree topologies have a root node, and all other nodes are connected which forming a hierarchy.
Hybrid Topology	Hybrid topology combines two or more topologies

► LAN Trainer:





► Overview of Network Types



1.3

► Overview

Types of Computer Networks (by their size / area/ geographical area).

1. PAN (Personal Area Network)
2. LAN(Local Area Network)
3. CAN(Campus Area Network)
4. MAN(Metropolitan Area Network)
5. WAN(Wide Area Network)

► PAN (Personal Area Network)

- A personal area network (PAN) is the interconnection of information technology devices within the **range of an individual person**, typically within a range of **10 meters(30 feet)**.
- PAN (Personal Area Network) is a computer network formed **around a person**.
- It generally consists of a computer, mobile, or personal digital assistant. PAN can be used for establishing communication among these personal devices for connecting to a digital network and the internet.

Two Types of PAN:

1. **Wireless Personal Area Network:** Wireless Personal Area Network is developed by simply using wireless technologies such as WiFi, Bluetooth. It is a low range network.
2. **Wired Personal Area Network:** Wired Personal Area Network is created by using the USB.



► PAN (Personal Area Network)

Characteristics of PAN:

- It is mostly personal devices network equipped within a limited area.
- Allows you to handle the interconnection of IT devices at the surrounding of a single user.
- PAN includes mobile devices, tablet, and laptop.
- It can be wirelessly connected to the internet called WPAN.
- Appliances use for PAN: cordless mice, keyboards, and Bluetooth systems.

Example:

Body Area Network: Body Area Network is a network that moves with a person. For example, a mobile network moves with a person. Suppose a person establishes a network connection and then creates a connection with another device to share the information.

► PAN (Personal Area Network)

Advantages of PAN:

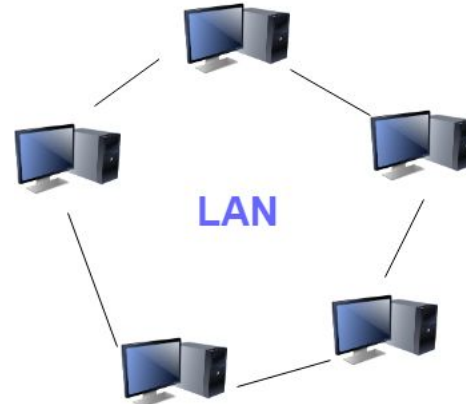
- PAN networks are relatively secure and safe
- It offers only short-range solution up to ten meters
- Strictly restricted to a small area

Disadvantages of PAN:

- It may establish a bad connection to other networks at the same radio bands.
- Distance limits.

► LAN (Local Area Network)

- A Local Area Network (LAN) is a group of computer and peripheral devices which are connected in a **limited area** such as school, laboratory, home, and office building.
- It is a widely useful network for **sharing resources** like files, printers, games, and other application. The simplest type of LAN network is to connect computers and a printer in someone's home or office.
- They are sharing a common connection that has **10-100 Mbps data transmission** speed and are connected by Ethernet cables, usually running on **high-speed internet connection**.



► LAN (Local Area Network)

Characteristics of LAN:

- ❖ It is a **private network**, so an outside regulatory body never controls it.
- ❖ LAN operates at a relatively higher speed compared to other WAN systems.
- ❖ There are various kinds of **media access control methods** like token ring and ethernet.
- ❖ It connects computers in a single building, block or campus(i.e a restricted geographical area).

Application/ Example:

- ❖ Connecting locally all the workstations in a building to let them communicate with each other locally without any internet access.
- ❖ Sharing common printer

► LAN (Local Area Network)

Advantages of LAN:

- ❖ Computer resources like hard-disks, DVD-ROM, and printers can share local area networks. This significantly reduces the cost of hardware purchases.
- ❖ Data of all network users can be stored on a single hard disk of the server computer.
- ❖ You can easily transfer data and messages over networked computers.
- ❖ It will be easy to manage data at only one place, which makes data more secure.
- ❖ Local Area Network offers the facility to share a single internet connection among all the LAN users.

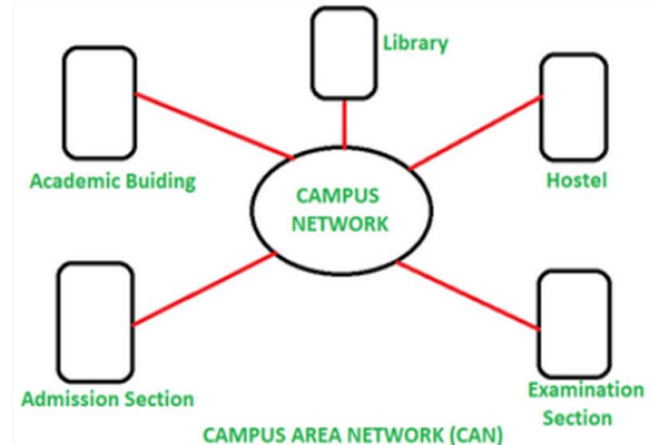
Disadvantages of LAN:

- ❖ LAN will indeed save cost because of shared computer resources, but the initial cost of installing Local Area Networks is quite high.
- ❖ The LAN admin can check personal data files of every LAN user, so it does not offer good privacy.
- ❖ Unauthorized users can access critical data of an organization in case LAN admin is not able to secure centralized data repository.
- ❖ Local Area Network requires a constant LAN administration as there are issues related to software setup and hardware failures

► CAN (Campus Area Network)

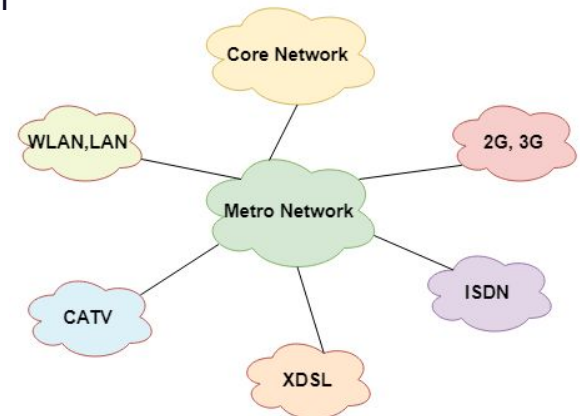
- ❖ Campus Area Network (CAN) is a group of **interconnected Local Area Networks (LAN)** within a limited geographical area like school campus, university campus, military bases, or organizational campuses and corporate buildings etc.
- ❖ This Campus Area Network also called as Corporate Area Network.
- ❖ This network is also referred as **Residential Network or ResNet** as it is only used by residents of specific campus only.
- ❖ **Owner:** University, Enterprise

Example: a university campus can be linked with a variety of campus buildings to connect all the academic departments.



► MAN (Metropolitan Area Network)

- A Metropolitan Area Network or MAN is consisting of a computer network across an entire city, college campus, or a small region.
- A metropolitan area network is a network that covers a **larger geographic area** by interconnecting a different LAN to form a larger network.
- The size of the MAN is **larger** than LANs and **smaller** than WANs (wide area networks), a MAN's covers the larger area of a city or town.
- Government agencies use MAN to connect to the citizens and private industries.
- Network in a City is called MAN (Metropolitan Area Network)
- **Owner:** Single Private Company or Public Company



► MAN (Metropolitan Area Network)

Characteristics of MAN:

- It mostly covers towns and cities in a maximum 50 km range
- Mostly used medium is optical fibers, cables
- Data rates adequate for distributed computing applications.

Application/ Example:

- Communication in Military
- Communication between banks in a city.

► MAN (Metropolitan Area Network)

Advantages of MAN:

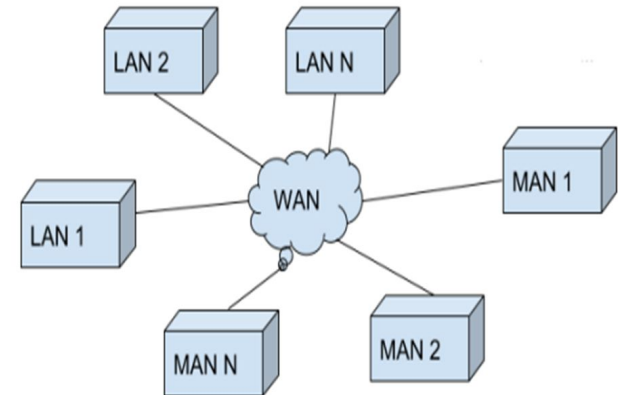
- It offers fast communication using high-speed carriers, like fiber optic cables.
- It provides excellent support for an extensive size network and greater access to WANs.
- The dual bus in MAN network provides support to transmit data in both directions concurrently.
- A MAN network mostly includes some areas of a city or an entire city.

Disadvantages:

- You need more cable to establish MAN connection from one place to another.
- In MAN network it is tough to make the system secure from hackers.

► WAN (Wide Area Network)

- A Wide Area Network is a network that extends over a large geographical area such as states or countries.
- A WAN is quite **bigger** network than the MAN.
- A WAN is **not limited** to a single location, but it spans over a large geographical area through a telephone line, fiber optic cable or satellite links.
- The internet is one of the biggest WAN in the world.
- A WAN is widely used in the field of Business, government, and education.



► WAN (Wide Area Network)

Characteristics of WAN:

- It covers large geographical areas(states, countries,continents).
- Communication medium used are satellites, public telephone networks connected by routers etc.

Application/ Example:

- Mobile Broadband: A 4G network is widely used across a region or country.
- Nepal Telecom: A telecom company is used to provide the internet services to the customers in hundreds of cities by connecting their home with fiber.

► WAN (Wide Area Network)

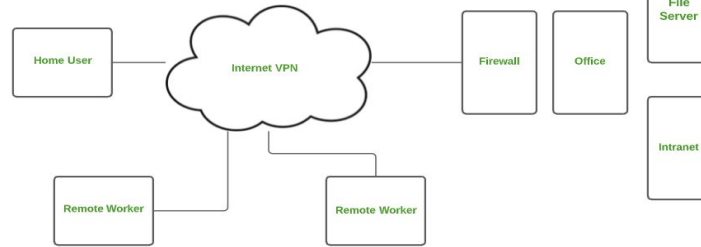
Advantages of WAN:

- WAN helps you to cover a larger geographical area. Therefore business offices situated at longer distances can easily communicate.
- Contains devices like mobile phones, laptop, tablet, computers, gaming consoles, etc.
- Shares software and resources with connecting workstations.
- Everyone in the internet use same data that resolves issue of some user may have to access the old information, while other access new.

Disadvantages:

- The initial setup cost of investment is very high.
- It is difficult to maintain the WAN network. You need skilled technicians and network administrators.
- There are more errors and issues because of the wide coverage and the use of different technologies.
- It requires more time to resolve issues because of the involvement of multiple technologies.
- Offers lower security compared to other types of network in computer

► Other Networks:



Virtual Private Network

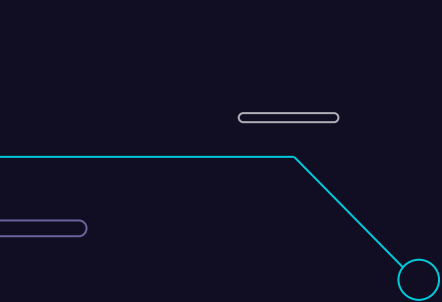
- A VPN is a **private network** which uses a **public network** to connect remote sites or users together. The VPN network uses “virtual” connections routed through the internet from the **enterprise’s private network** or a third-party VPN service to the remote site.
- It is a **free or paid** service that keeps your web browsing secure and private over public WiFi hotspots.

Internetwork:

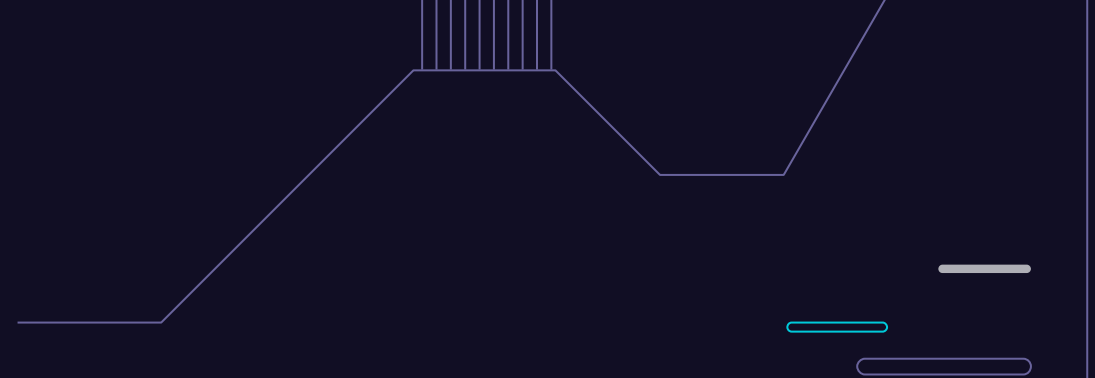
- An internetwork is defined as **two or more** computer network LANs or WAN or **computer network segments** are connected using devices, and they are configured by a **local addressing scheme**. This process is known as internetworking.
- An interconnection between **public, private, commercial, industrial, or government** computer networks can also be defined as internetworking.
- An internetworking uses the **internet protocol**.
- The reference model used for internetworking is Open System Interconnection(OSI).

► Summary & Comparison:

Parameter	PAN	LAN	CAN	MAN	WAN
Full Form	Personal Area Network	Local Area Network	Campus Area Network	Metropolitan Area Network	Wide Area Network
Technology	Bluetooth, IrDA, Zigbee	Ethernet and Wifi	Ethernet	FDDI,CDDI,ATM	Leased line ,Dial-Up
Range	1-10 meter (Max)	Upto 2KM	1-5 KM	5-50KM	Above 50 KM
Transmission Speed	Very High	Very High	High	Average	Low
Area	Within in a room	Within in office ,buildings	Within university ,corporate offices	Within City	Within in countries.
Ownership	Private	Private	Private	Private or Public	Private or Public
Maintenance	Very Easy	Easy	Moderate	Difficult	Very Difficult
Error Rate and Cost	Very low	Low	Moderate	High	Very High



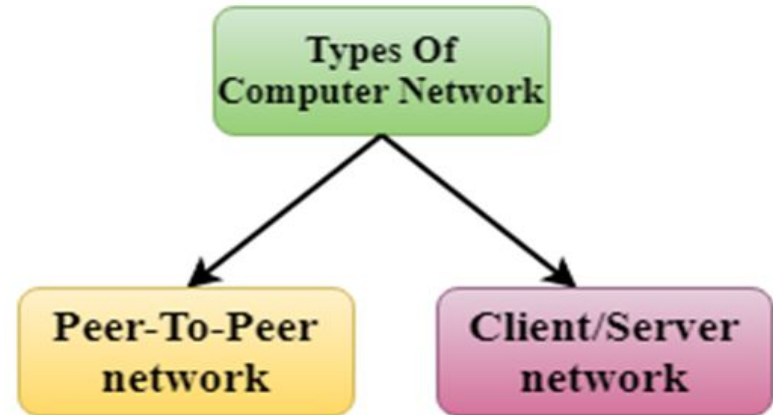
Networking ► Types (Architecture)



1.4

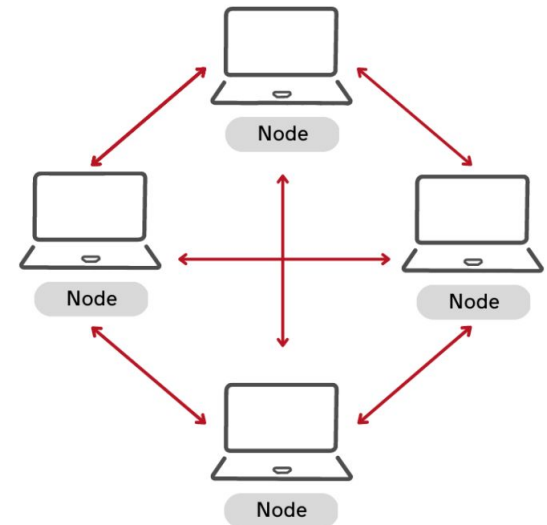
► Networking Architecture:

- Computer Network Architecture is defined as the **physical and logical design** of the software, hardware, protocols, and media of the transmission of data.
- Architecture also defines how the computers should get connected to get the **maximum advantages** of a computer network such as better response time, security, scalability etc.
- Generally two types:
 - Peer-To-Peer network
 - Client/Server network



► Peer to Peer Architecture (P2P):

- Peer-To-Peer network is a network in which all the computers are linked together with **equal privilege and responsibilities** for processing the data.
- Files can be shares **directly** between systems on the network without the need of central server.
- Each computer on p2p network becomes a file server as well as a client.
- Once connected to N/w, P2P software **allows** you to **search** for files on other clients computers.
- Peer-To-Peer network is useful for **small** environments, usually up to 10 computers.



► Peer to Peer Architecture (P2P):

Advantages:

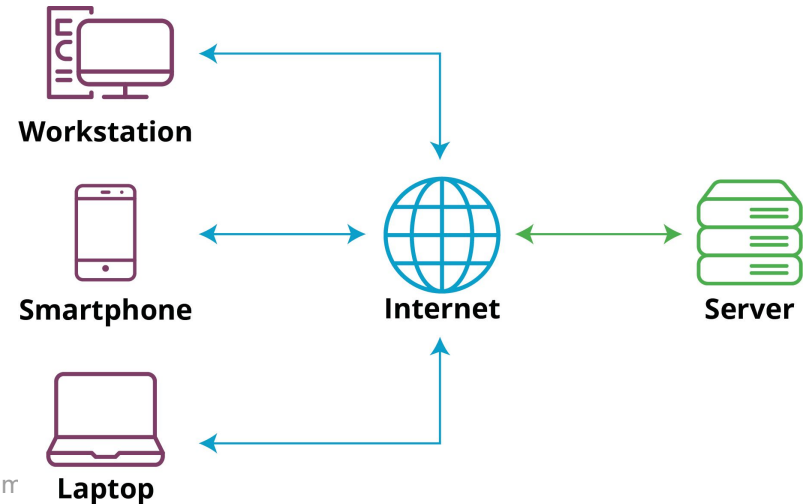
- **Less costly** is no central server that has to take the backup.
- In case of a computer failure all other computers in the network are **not affected** and they will continue to work as same as before the failure.
- **Installation** of peer to peer architecture is quite **easy** as each computer manages itself.

Disadvantages:

- **Each computer** has to take the backup **rather** than a central computer and the security measures are to be taken by **all the computers separately**.
- **Scalability** is an issue in a Peer to Peer Architecture as connecting each computer to every computer is a headache on.

► Client-Server Architecture:

- Client/Server network is a network model designed for the end users called **clients**, to access the resources such as songs, video, etc. from a central computer known as **Server**.
- The central controller is known as a server while all other computers in the network are called clients.
- A server performs all the major operations such as security and network management.
- **Examples of the Client-Server Model :**
Email, World Wide Web, etc.

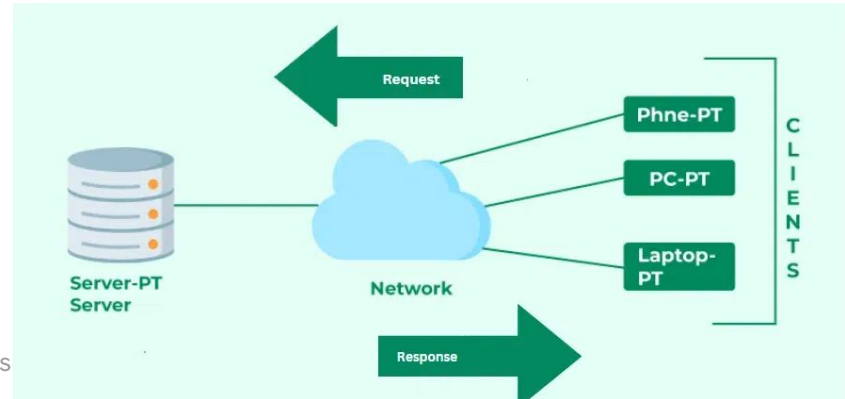


► Client-Server Architecture:

Server:

- Provides services for clients.
- Also known as a host computer.
- Controls access h/w and s/w on the network.
- Provide storage area for programs, data and information.
- Provide access to stored data.

Example: Application Server, Message Server, Proxy Server, Database Server, Web Server etc.

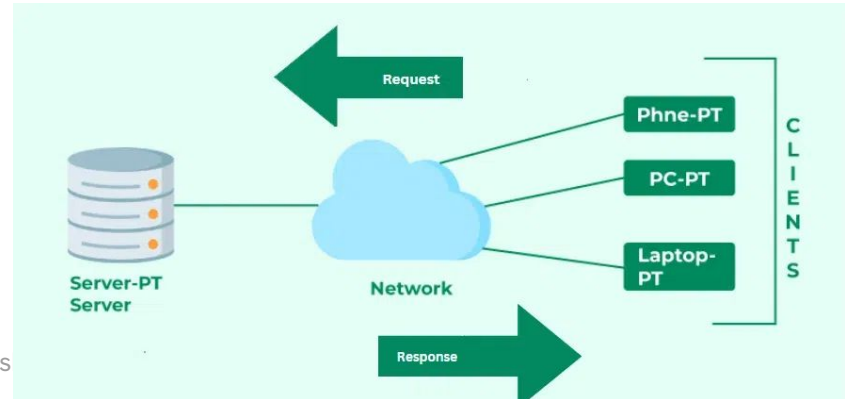


► Client-Server Architecture:

Clients:

- Also known as user or computer or end user.
- Request the resources from the server.
- Independent from other clients
- Gets the file from server
- Sends file to the server
- Delete, copy ,Rename the files on the server.

Example: Mobile Phones, Browser, Application etc



► Client-Server Architecture :

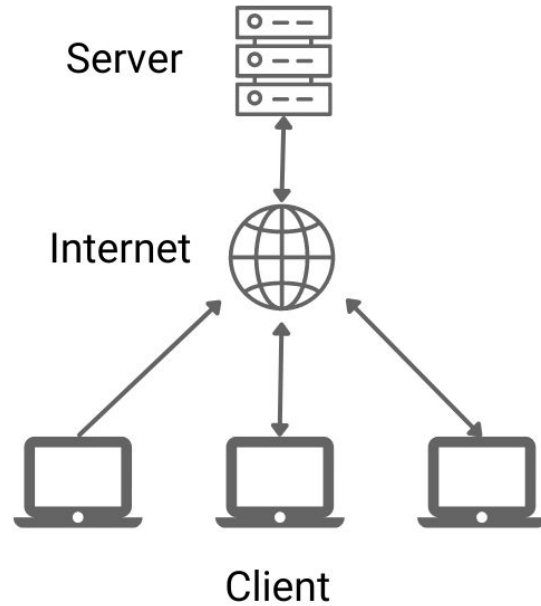
Advantages:

- **Data backup is easy and cost effective** as there is no need to manage the backup on each computer.
- **Performance is better** as the response time is greatly improves because the server is more powerful computer than the other computers in the network.
- **Security is better** as unauthorized access are denied by server computer and all the data goes through the server.
- **Scalability is not an issue** in this Architecture as large number of computers can be connected with server.

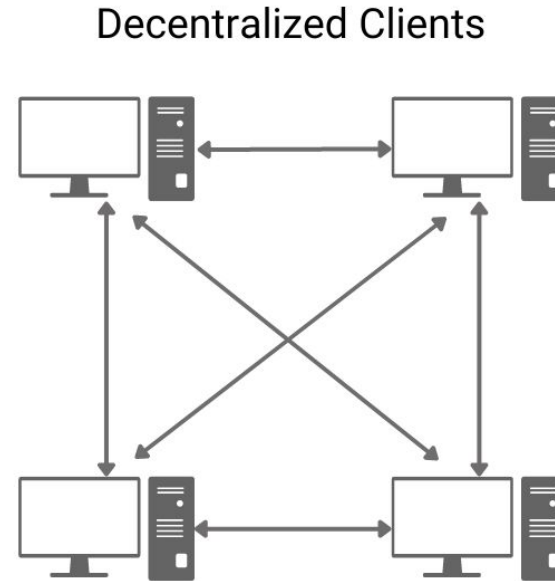
Disadvantages:

- Client/Server network is expensive as it requires the server with large memory.
- A server has a Network Operating System(NOS) to provide the resources to the clients, but the cost of NOS is very high.
- It requires a dedicated network administrator to manage all the resources.

► Client-Server vs P2P Architecture :



Client Server Architecture



Peer to Peer Architecture

► Client-Server vs P2P Architecture :

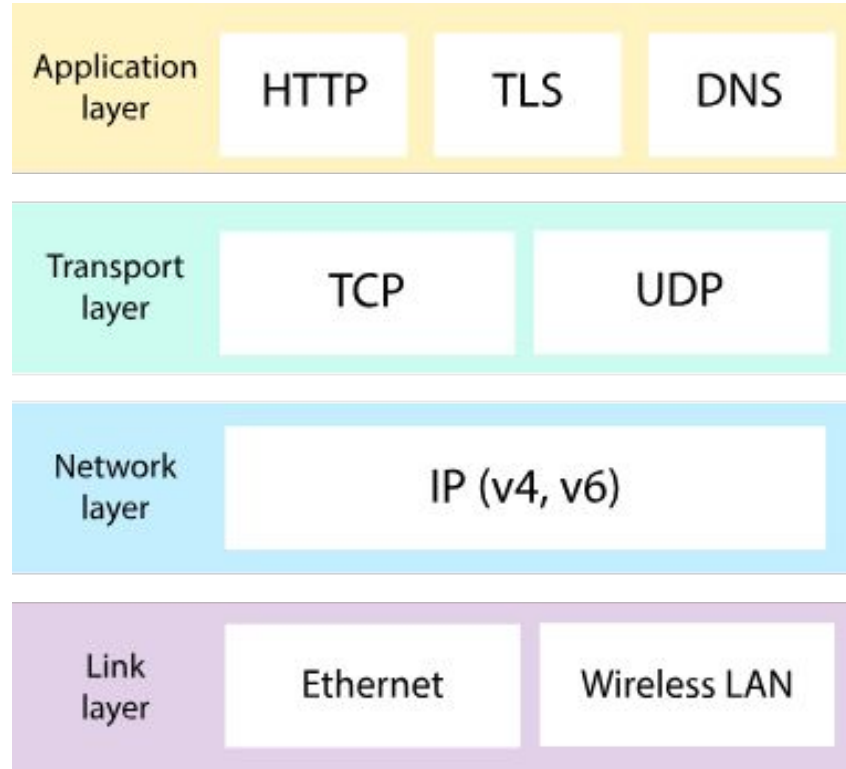
Client-Server Architecture	Peer-to-Peer Architecture
A clear separation between clients and servers.	No differentiation between clients and servers.
Data is provided only in response to a request.	Peers have the authority to request as well as provide a service.
Centralized data management.	It has own data and applications.
Purpose is to store and exchange information.	Goal is to maintain connections among peers.
Suitable for small as well as large networks.	Suitable for less number of users or devices.



Overview of ► Protocols and Standards

1.5

► Protocols:



► Overview of Protocols and Standards:

Protocol:

- A protocol is a set of rules that governs(control) data communications.
- A protocol defines **what** is communicated, **how** is communicated, and **when** it is communicated between connected devices, **regardless of differences** in their internal process, structure or design..

The **key elements** of a protocol are :

1. Syntax
2. Semantics
3. Timing

► Overview of Protocols and Standards:

Syntax:

- Structure or format of the data, meaning the **order in which they are packaged by the sender and opened up by the receiver (Presented)**.
- Syntax should be **same** in sender and receiver for to communicate.
- **Example:** A simple protocol might expect the **first** 8 bits of data to be the address of the sender, the **second** 8 bits to be the address of the receiver, and the **rest** of the stream to be the message itself.
- The data order is also applied to the order of bits when they are **stored or transmitted**.

► Overview of Protocols and Standards:

Semantics:

- Interprets the **meaning of the each section of bits** in a network protocol.
- It allows sender and receiver to **interpret the pieces correctly**. How is a particular pattern to be interpreted, and what action is to be taken based on that interpretation?
- **Example:** Does an address identify route to be taken or final destination of the message.

► Overview of Protocols and Standards:

Timing:

- Timing refers to **two characteristics**: **when** data should be sent and **how** fast it should be sent.
- **Speed** at which data should be **sent** or speed at which it is being **received**.
- **Example**: If a sender produces data at 100 Mbps but the receiver can process data at only 1 Mbps, the transmission will overload the receiver and some data will be lost.

► Standards:

Standards are guidelines that explain to all IT stakeholders -from device manufacturers to software programmer and network admins on how a particular protocol should operate.

- Standards provide **guidelines** to manufacturers, vendors, government agencies, and other service providers to ensure the kind of **interconnectivity** necessary in today's marketplace and in **international communications**.
- Standards are essential in creating and maintaining an **open and competitive** market for equipment manufacturers and in guaranteeing.
- Data communication standards fall into **two categories**:
 - **de facto** (meaning "by fact" or "by convention")
 - **de jure** (meaning "by law" or "by regulation").
-

► Two Categories of Standards:

De facto:

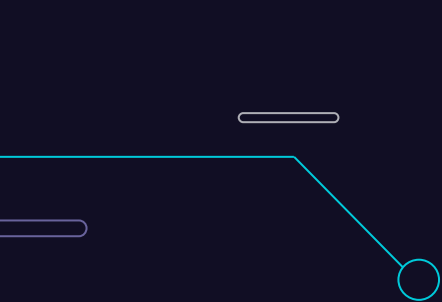
- Standards that **have not been approved** by an organized body **but have been adopted** as standards through widespread use are de facto standards. De facto standards are **often established originally** by manufacturers who seek to define the functionality of a **new product or technology**.

De jure:

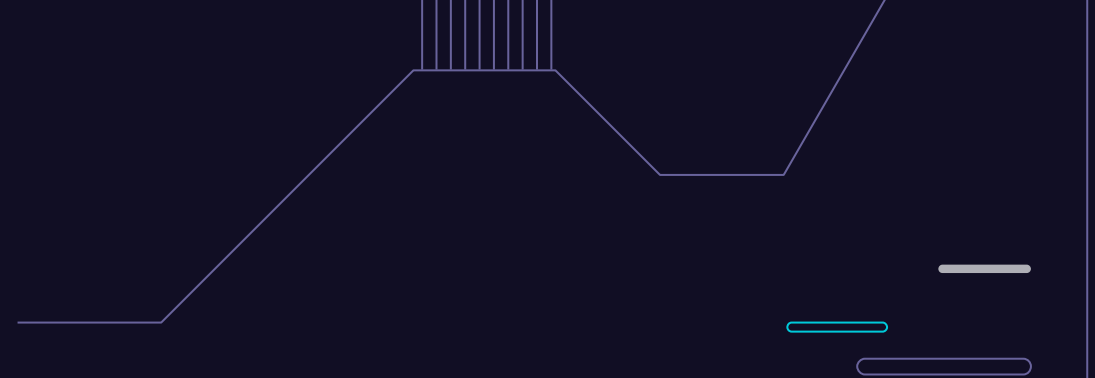
- Those standards by **law or by regulation**. These are the standards recognized officially by an Organization.

Some standards are:

- International Organization for Standardization (ISO)
- International Telecommunication Union-Telecommunication Standards Sector (ITU-T)
- American National Standards Institute (ANSI)
- Institute of Electrical and Electronics Engineers (IEEE)
- Electronic Industries Association (EIA)



▶ OSI Reference Model



1.6

► OSI Reference Model:

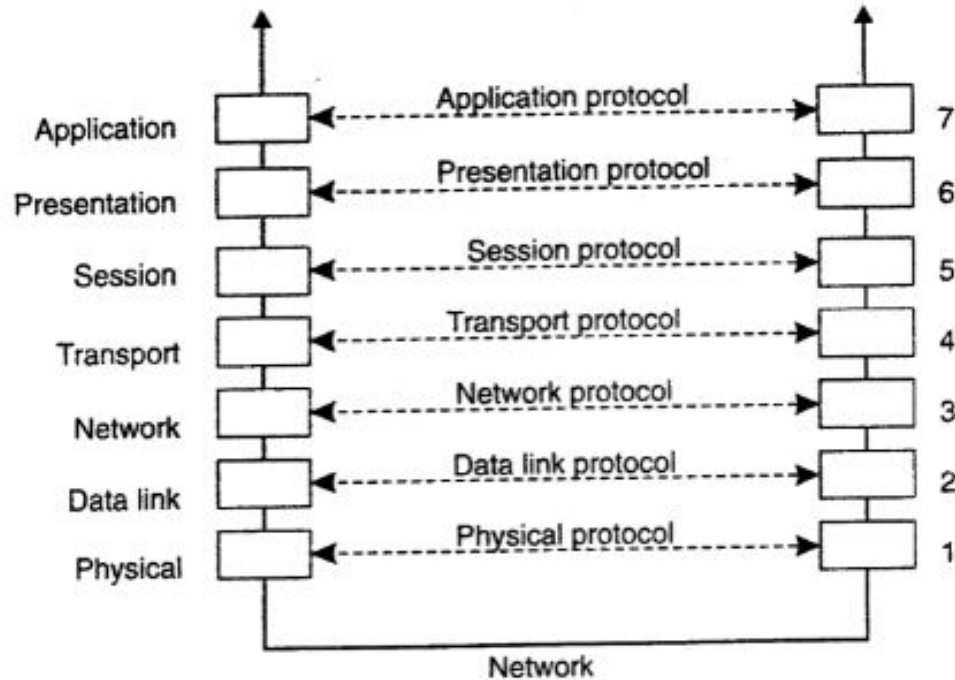
Issue: There are many users who use computer network and are located all over the world. To ensure national and worldwide data communication.

- **OSI- Opens System Interconnections.**

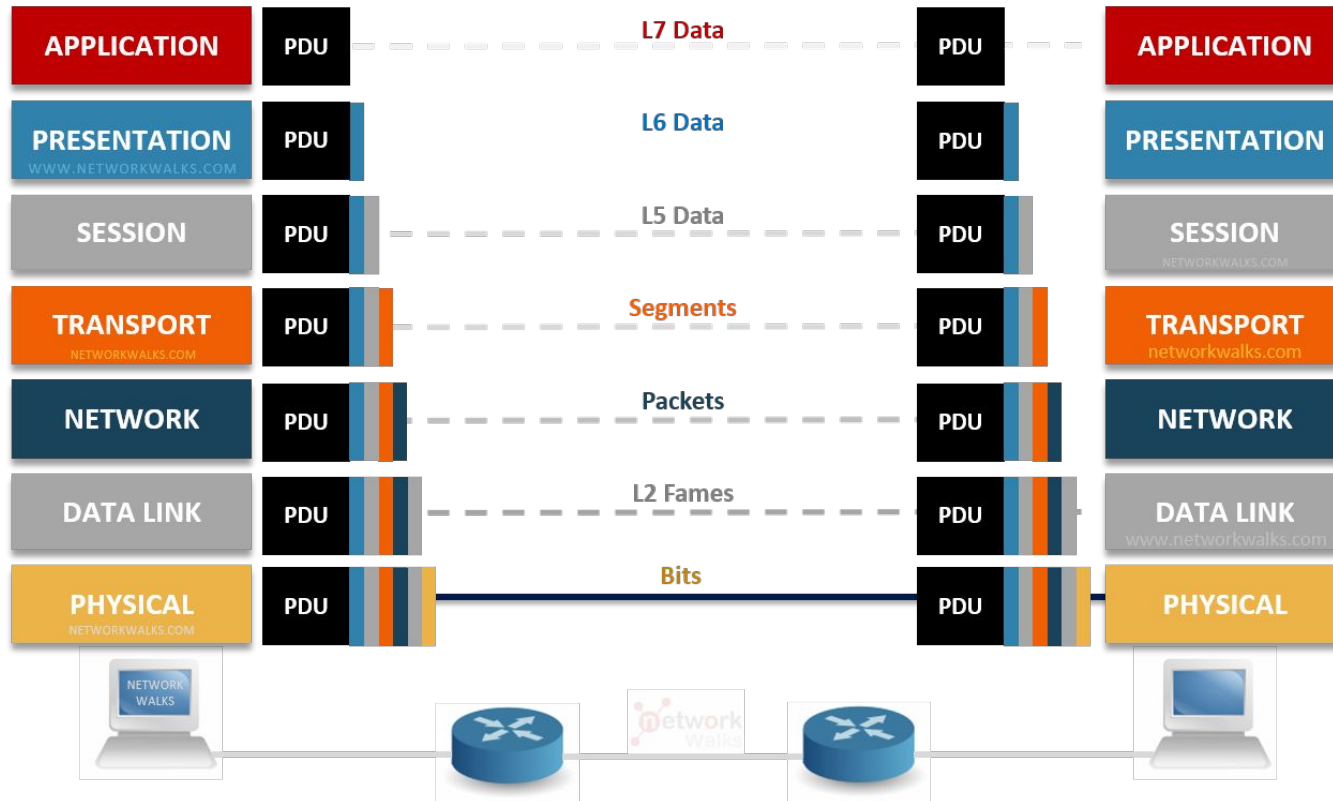
- It was proposed by ISO (International Organization for Standards) in 1984 **after the TCP/IP model**.
- Starts developing in late 1970s.
- Approved by 1984.
- The term "Open" in Open System Interconnections denotes **"to communicate with any 2 systems"**.
- There are **7 layers** in OSI Reference model.
- It is also called **OSI layered architecture / OSI Protocol architecture**.
- The process of breaking up the functions or tasks of networking into layers **reduces complexity**.
- Each layer provides a **service to the layer above it** in the protocol specification.
- Each layer communicates with the **same layer's software or hardware** on other computers.

► OSI Reference Model:

Please
Do
Not
Throw
Sushi and
Pizza
Away.
(PDNTSPA)



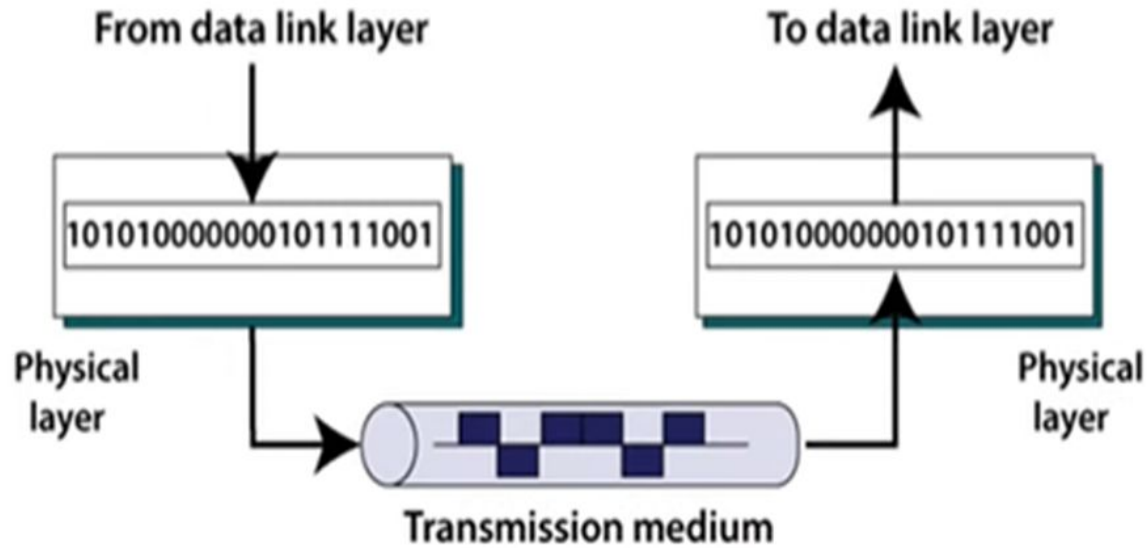
► Data Forms in Each Layer:



► Physical Layer:

- In the OSI model, the physical layer is the **lowest layer**, and it deals with the **actual physical transmission** of data over a **physical medium**, such as copper wires, optical fibres, or wireless radio waves.
- It defines the **hardware characteristics and electrical/optical specifications** of the transmission medium (i.e voltage, data rates etc).
- **Examples** of devices at the OSI physical layer include Hub, Repeater, Modem, Cables.
- It deals with the mechanical and electrical specifications of the interface and transmission medium. It **converts** the digital/analog bits into electrical signal or optical signals.
- It activates, maintains and deactivates the physical connection.
- **Data encoding** is also done in this layer.
- **PDU : Bits**

► Physical Layer:



► Physical Layer:

Functions of the Physical Layer:

- **Bit synchronization:** The physical layer provides the synchronization of the bits by providing a **clock**. This clock controls both sender and receiver thus providing **synchronization at the bit level**.
- **Bit rate control:** The Physical layer also defines the **transmission rate** i.e. the number of bits sent per second.
- **Physical topologies:** Physical layer specifies how the different, devices/nodes are arranged in a network i.e. bus, star, or mesh topology.
- **Transmission mode:** Physical layer also defines how the data flows between the two connected devices. The various transmission modes possible are **Simplex, half-duplex and full-duplex**.

► Data Link Layer:

- The data link layer is responsible for the **node-to-node delivery** of the message.
- The **main function** of this layer is to make sure data transfer is error-free from one node to another, over the physical layer.
- When a packet arrives in a network, it is the responsibility of the DLL to transmit it to the Host using its **MAC address**.
- The Data Link Layer is divided into **two sublayers**:
 - Logical Link Control (LLC)
 - Media Access Control (MAC)
- The MAC sub-layer is responsible for data encapsulation and accessing media.
- **PDU : Frame (Ethernet Frame)**

► Data Link Layer:

1. Logical Link Control Layer

- It is responsible for transferring the **packets to the Network layer of the receiver** that is receiving.
- It identifies the **address** of the network layer protocol from the **header**.
- It also provides flow control.

2. Media Access Control Layer

- A Media access control layer is a link between the **Logical Link Control** layer and the network's **physical layer**.
- It is used for transferring the packets over the network.



- **Encapsulation:**

- In encapsulation IP packet received from network layers are added with header and trailer.

- **Header:**

- Contains MAC address of Sender and Receiver.

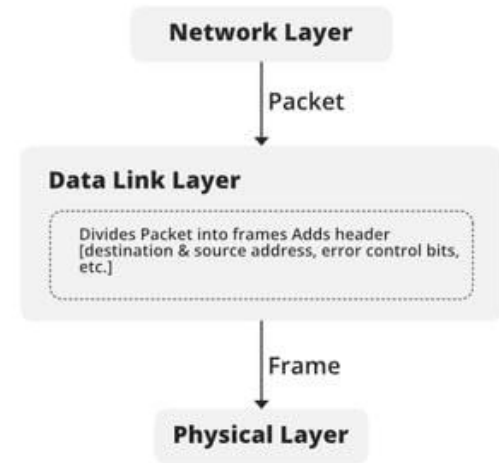
- **Trailer:**

- 4 bytes of **error checking data** used to detect error in received Ethernet frame.

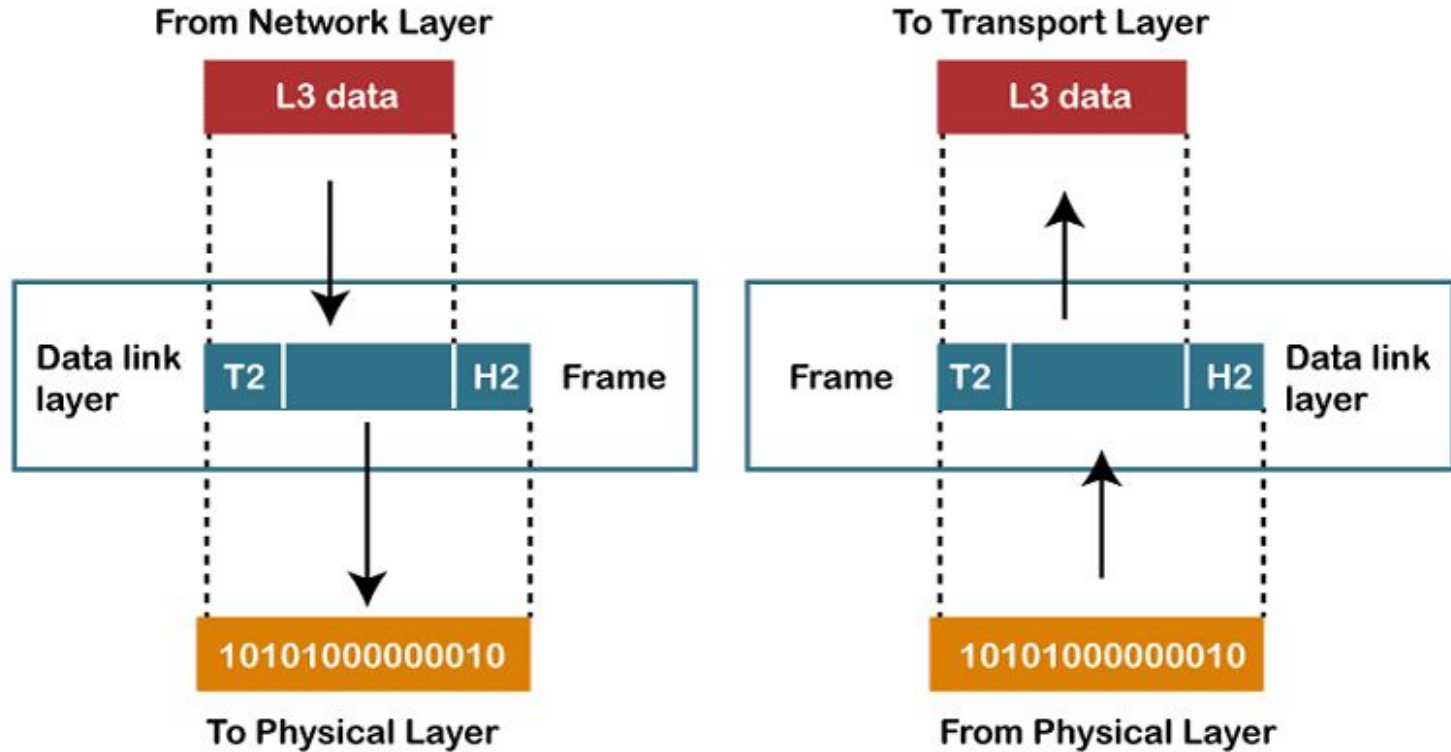
► Data Link Layer:

Functions of the Data Link Layer:

- **Framing:** Framing is a **function** of the data link layer. It provides a way for a sender to **transmit a set of bits** that are meaningful to the receiver. This can be accomplished by attaching special bit patterns to the **beginning and end of the frame**.
- **Physical addressing:** After creating frames, the Data link layer adds physical addresses (**MAC addresses**) of the sender and/or receiver in the **header of each frame**.
- **Error control:** The data link layer provides the mechanism of error control in which it detects and retransmits **damaged or lost** frames.
- **Flow Control:** The data rate must be constant on both sides else the data may get **corrupted** thus, flow control **coordinates the amount of data** that can be sent before receiving an acknowledgment.



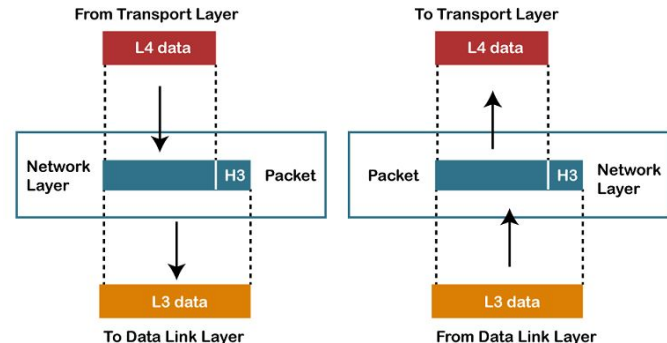
► Data Link Layer:



► Network Layer:

- The transport layer passes the TCP segments or UDP datagram to the network layers.
- Then NL accept the transport layer **segments** and add **logical addressing** and form a **packet**.
- Responsible for **moving packets** from source to destination as it determines the best path to move data from source to the destination based on the **network conditions**, the **priority of service**, and **other factors**.
- It divides the outgoing messages into packets and assembles the incoming packets into messages for higher levels.

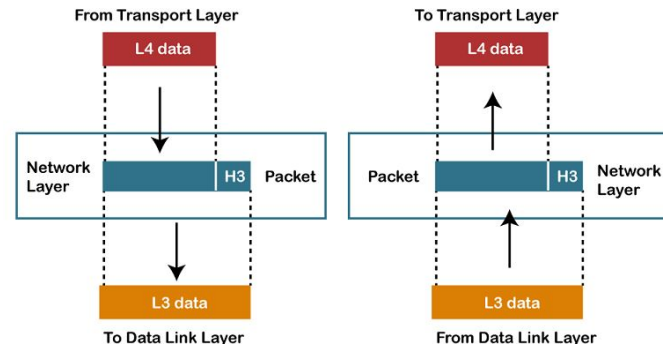
- **PDU : Packets**



► Network Layer:

Functions of the Network Layer :

- **Routing:** The network layer protocols determine which route is suitable from source to destination. This function of the network layer is known as routing.
- **Logical Addressing:** To identify each device inter-network uniquely, the network layer defines an addressing scheme. The sender & receiver's IP addresses are placed in the header by the network layer. Such an address distinguishes each device uniquely and universally.
- **Packetizing:** A Network Layer receives the data from the upper layer and converts them into packets. This process is known as Packetizing. It is achieved by internet protocol (IP).



► Transport Layer:

- Known as “Heart of OSI Model”.
- The Transport layer ensures that messages are transmitted **in the order** in which they are sent and there is **no duplication** of data.
- The main responsibility of the transport layer is to transfer the data **completely**.
- It receives the data from the upper layer and converts them into **smaller units** known as **segments**.
- This layer can be termed as an **end-to-end layer** as it provides a point-to-point connection between source and destination to deliver the data reliably.
- When the message reaches the transport layers, one transport layer protocols TCP or UDP is selected.
- Normally, OSI transport layer is **connection oriented** . i.e. TCP is selected.
- Data in the Transport Layer is called **Segments**.

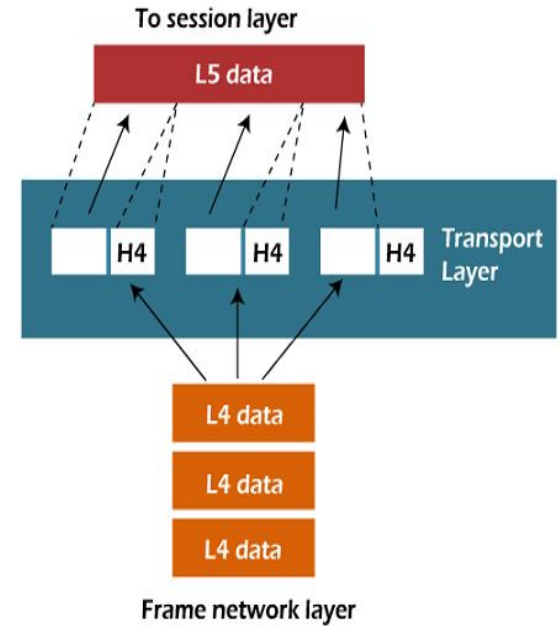
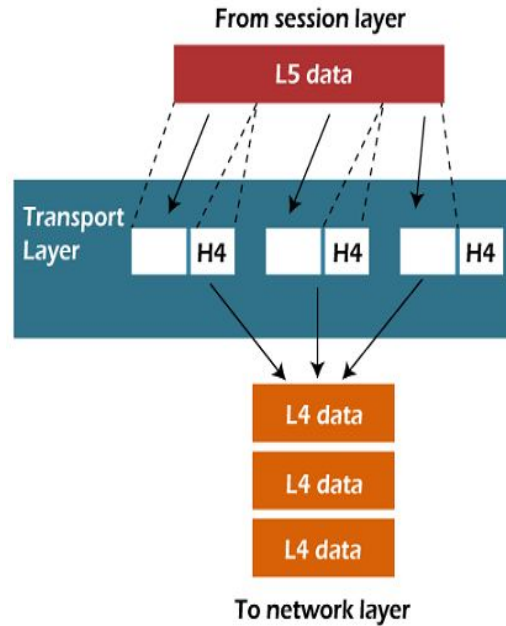
► Transport Layer:

TCP

- Supports segmentation
- Connection oriented
- Reliable
- TCP segment

UDP

- Does not support segmentation
- Connection less
- Unreliable
- UDP datagram



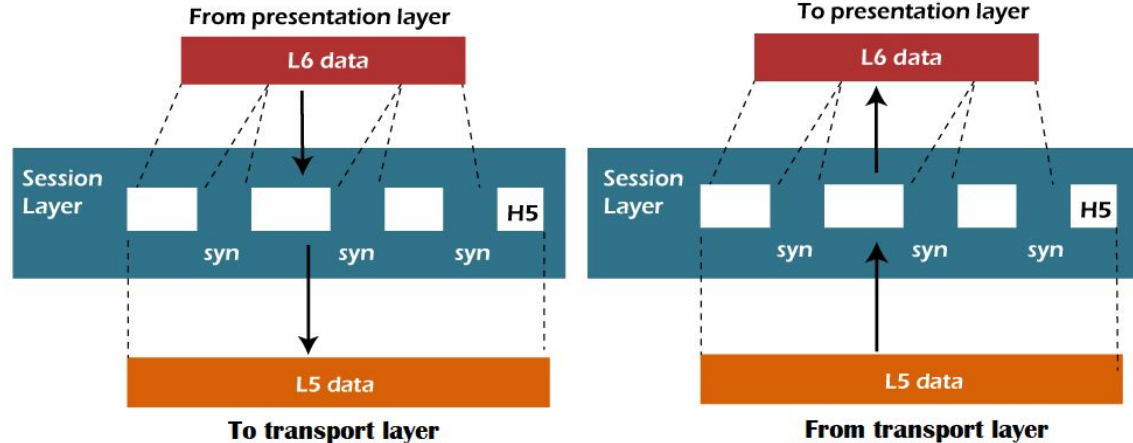
► Transport Layer:

Functions of the Transport Layer

- **Segmentation and Reassembly:** This layer accepts the message from the (session) layer, and breaks the message into smaller units. Each of the **segments** produced has a header associated with it. The transport layer at the destination station reassembles the message.
- **Service Point Addressing:** To deliver the message to the correct process, the transport layer header includes a type of address called **service point address** or port address. Thus by specifying this address, the transport layer makes sure that the message is delivered to the correct process.
- **Connection control:** Transport layer provides two services **Connection-oriented** service and **connectionless** service.

► Session Layer:

- Creates **communication channel** and controlling ports and session.
- It establishes, maintains, and ends a **session**.
- It is also responsible for **dialog control** and **coordinates communication** between the system.
- Session layer enables **two systems to enter** into a dialog/Communication.



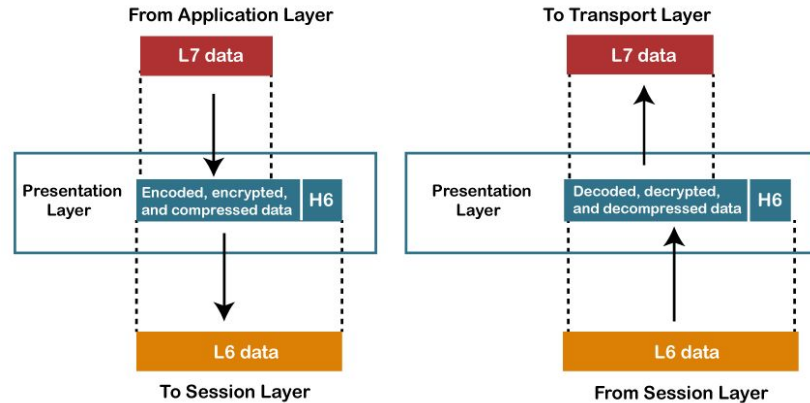
► Session Layer:

Functions of the Session Layer:

- **Session establishment, maintenance, and termination:** The layer allows the two processes to establish, use, and terminate a connection.
- **Synchronization:** This layer allows a process to add checkpoints that are considered synchronization points in the data. These synchronization points help to identify the error so that the data is re-synchronized properly, and ends of the messages are not cut prematurely and data loss is avoided.
- **Dialog Controller:** The session layer allows two systems to start communication with each other in half-duplex or full-duplex.

► Presentation Layer:

- The presentation layer is also called the **Translation layer**. The data from the application layer is **extracted** here and **manipulated** as per the required **format** to transmit over the network.
- Presentation Layer is responsible for to Translate, Compress, Encrypt and Decrypt data.
- A Presentation layer is mainly concerned with the **syntax and semantics** of the information exchanged between the two systems.



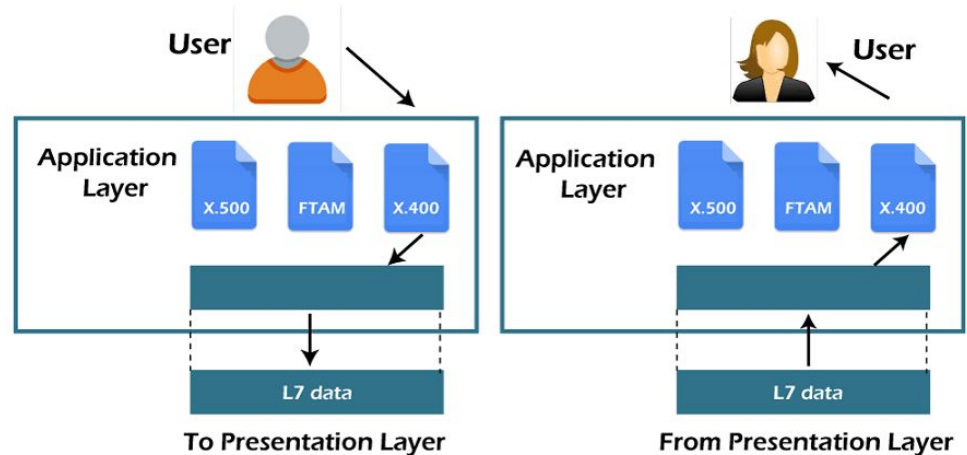
► Presentation Layer:

Functions of the Presentation Layer:

- **Translation:** For example, ASCII to EBCDIC.
- **Encryption/ Decryption:** Data encryption translates the data into another form or code. The encrypted data is known as the ciphertext and the decrypted data is known as plain text. A key value is used for encrypting as well as decrypting data.
- **Compression:** Reduces the number of bits that need to be transmitted on the network.

► Application Layer:

- Application layer allows **access to** network resources.
- In the application layer, an application, such as an internet browser, gets the data and a user can then interact with it.
- **Highest level** of OSI layers.
- Application-layer helps to **identify** communication partners, **determining** resource availability, and **synchronizing** communication.
- It allows users to **log on** to a remote host.
- This layer provides various e-mail services.
- This is also called **closest layer** from user.



► Application Layer:

Functions of Application layer:

- **File transfer, access, and management (FTAM):** An application layer allows a user to access the files in a remote computer, to retrieve the files from a computer and to manage the files in a remote computer.
- **Mail services:** An application layer provides the facility for email forwarding and storage.
- **Directory services:** An application provides the distributed database sources and is used to provide that global information about various objects.

► Data Flow in OSI



Data flow in OSI Layers explained with an example:

📧 Sending an email to a friend involves passing through various layers of communication:

- ◆ **Application Layer:** Email data is sent from your email software to the communication stack. The data passes through this layer to the next one.
- ◆ **Presentation Layer:** Data compression takes place in this layer to optimize the email's size.
- ◆ **Session Layer:** Communication is initialized at this layer to establish a connection.
- ◆ **Transportation Layer:** The data is segmented into smaller units for efficient transmission.
- ◆ **Network Layer:** The segmented data is further broken down into packets with routing information.
- ◆ **Data Link Layer:** Packets are encapsulated into frames for transmission over the network.
- ◆ **Physical Layer:** The frames are converted into binary signals (0s and 1s) suitable for transmission over physical media like cables.

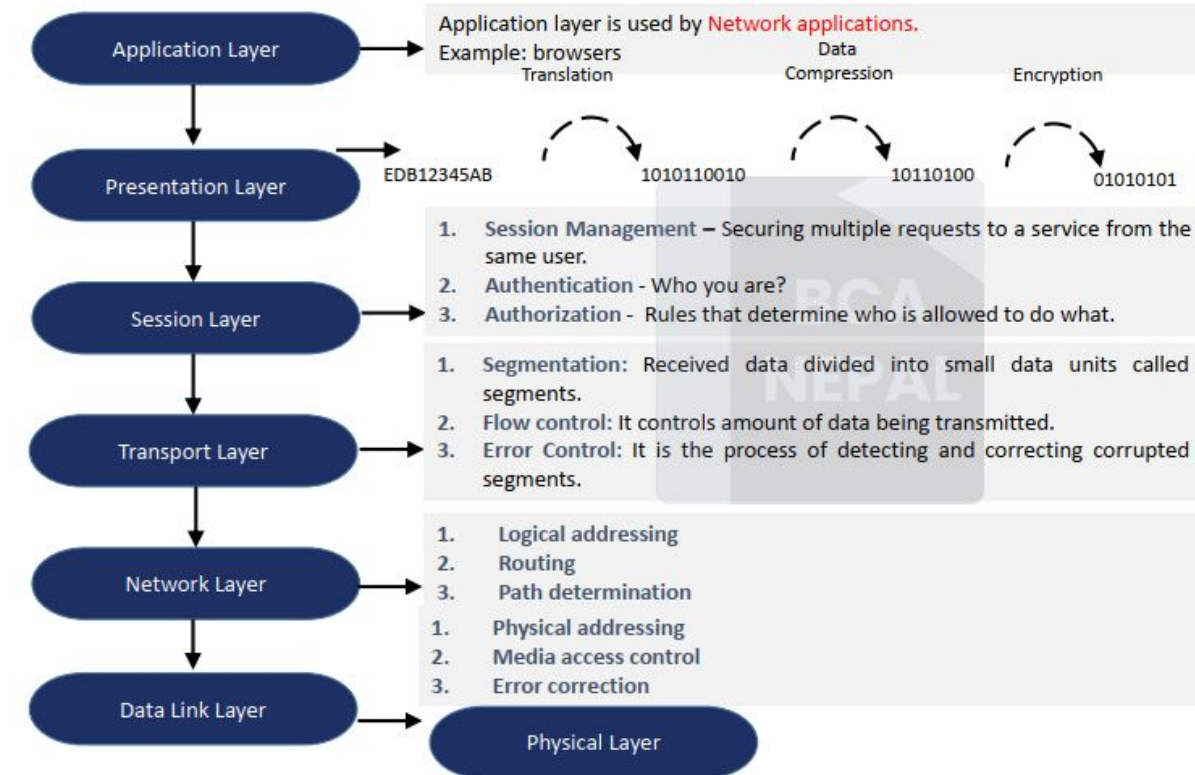
Link For Details:

📧 When the email reaches your friend:

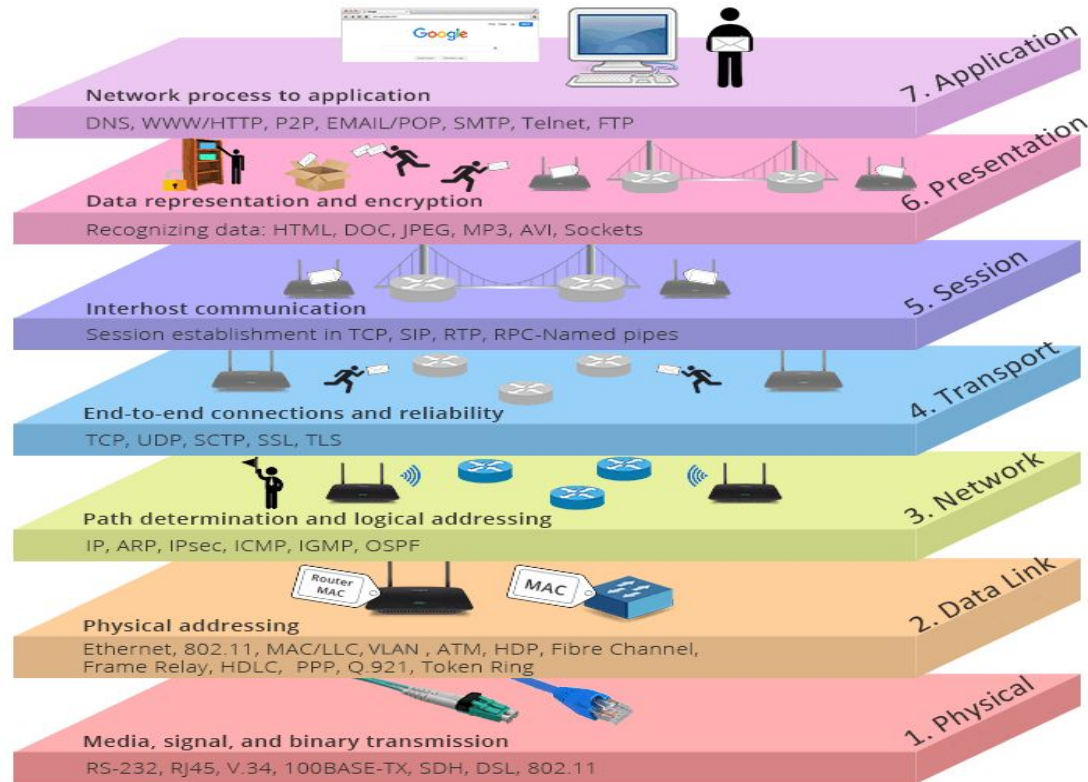
- The data goes through the layers in the opposite order to be reconstructed and presented:

- ◆ **Physical Layer:** The binary signals are converted back into frames.
- ◆ **Data Link Layer:** Frames are reassembled into packets.
- ◆ **Network Layer:** The packets are combined to form the original segmented data.
- ◆ **Transportation Layer:** Segmented data is assembled into its complete form.
- ◆ **Session Layer:** The communication session is concluded.
- ◆ **Presentation Layer:** Any data decompression or format conversion occurs here.
- ◆ **Application Layer:** The data is delivered to the email software in a human-readable format for your friend to read the email.

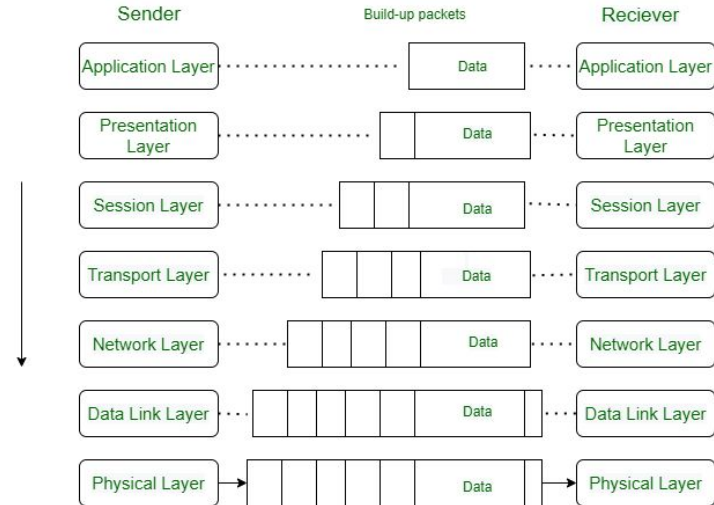
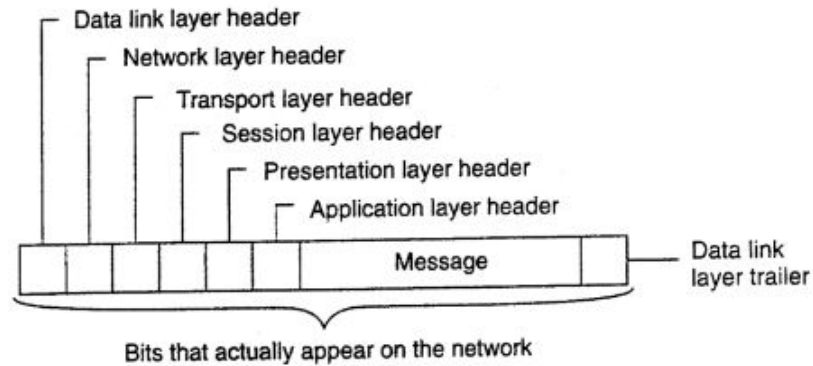
► OSI Reference Model:



► OSI Reference Model:



► A typical message as it appears in the network:



When the message arrives at machine 2, it is passed upward, with each layer stripping off and examining its own header. Finally, the message arrives at the receiver, process B, which may reply to it using the reverse path. The information in the layer n header is used for the layer n protocol.

► OSI Model:

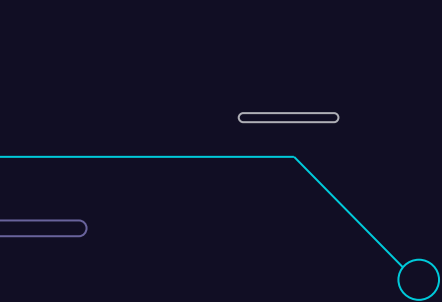
Advantages:

- OSI model distinguishes well between the services, interfaces and protocols.
- Protocols of OSI model are very well hidden.
- Protocols can be replaced by new protocols as technology changes.
- Supports connection oriented services as well as connectionless service.

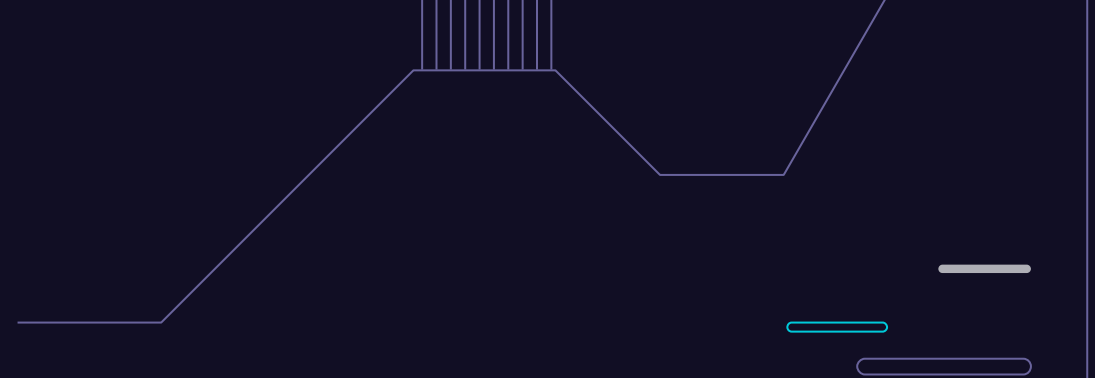
Disadvantages:

- Model was devised before the invention of protocols.
- Fitting of protocols is tedious task.
- It is just used as a reference model.

Layer No	Layer Name	Responsibility	Information Form (Data Unit)	Device	Protocol
7	<u>Application Layer</u>	Helps in identifying the client and synchronizing communication.	Message	-	SMTP, HTTP, FTP, POP3, SNMP, DNS etc
6	<u>Presentation Layer</u>	Data from the application layer is extracted and manipulated in the required format for transmission.	Message	-	MPEG,ASCH,SSL,TLS,JPEG,GIF
5	<u>Session Layer</u>	Establishes Connection, Maintenance, Ensures Authentication and Ensures security.	Message (or encrypted message)	Gateway	NetBIOS, SAP
4	<u>Transport Layer</u>	Take Service from Network Layer and provide it to the Application Layer.	Segment	Firewall	TCP, UDP
3	<u>Network Layer</u>	Transmission of data from one host to another, located in different networks.	Packet	Router	IPV4, IPV6, ICMP, IPSEC, ARP, MPLS.
2	<u>Data Link Layer</u>	Node to Node Delivery of Message.	Frame	Switch, Bridge	RAPA, PPP, Frame Relay, ATM, etc.
1	<u>Physical Layer</u>	Establishing Physical Connections between Devices.	Bits	Hub, Repeater, Modem, Cables	RS232, 100BaseTX, ISDN



TCP/IP Model & ► Its Comparison with OSI Model



1.7

► Brief History of TCP/IP Model:

- TCP/IP protocol suite was developed **before** the OSI model.
- TCP/IP is a set of protocols developed to allow **co-operating computers** to share resources across a network.
- In 1969 the Defense Advanced Research Projects Agency (**DARPA**) funded a research and development project to create an **experimental packet switching network**. This network is called **ARPANET**.
- In 1975 the ARPANET was **converted** from an experimental network to an **operational network**, and the responsibility for administering the network was given to the **Defense Communication Agency (DCA)**.
- The TCP/IP protocols were adopted as **Military Standards (MIL STD)** in 1983, and all hosts connected to the network were **required to convert** to the new protocols.
- DARPA funded to implement **TCP/IP** in Berkely Unix.
- In 1983, the old ARPANET was divided into **MILNET** and smaller **ARPANET**. The Internet was used to refer to the **entire network**; MILNET and ARPANET.

► TCP/IP Model Preface:

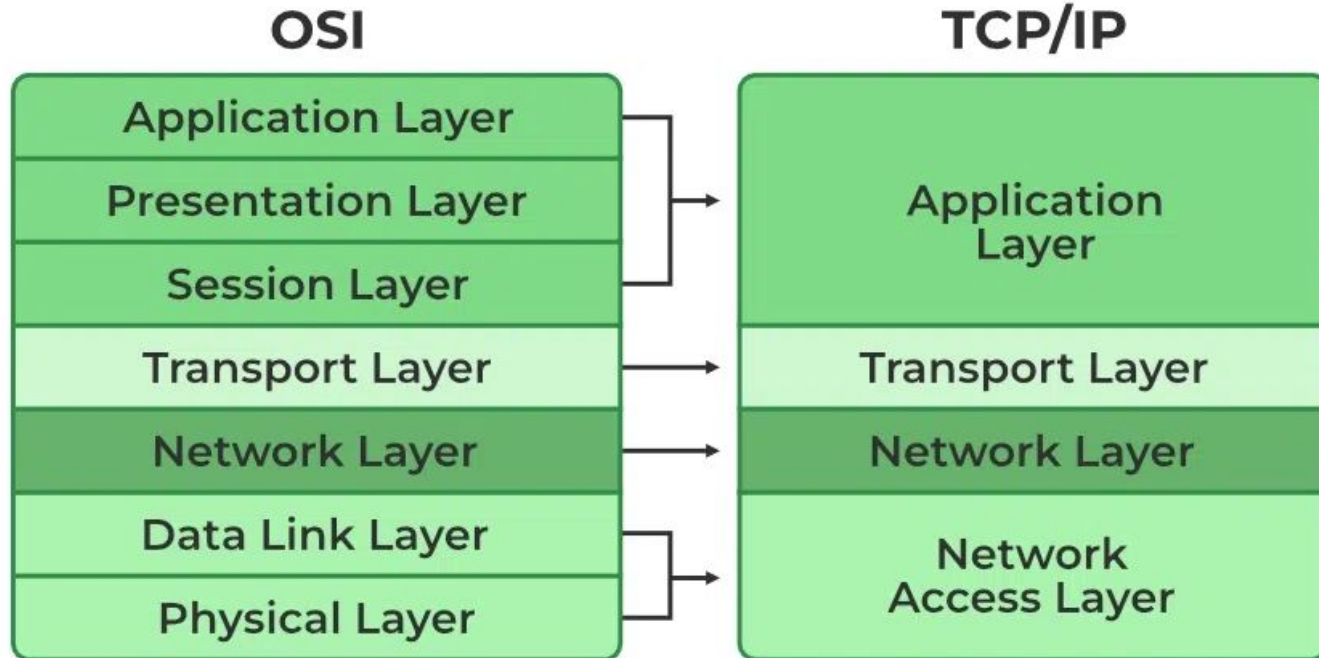
The features that **stood out** during the research, which led to **making** the TCP/IP reference model were:

- Support for a **flexible architecture**. Adding more machines to a network was easy.
- The network was robust, and **connections remained intact** until the source and destination machines were functioning.
- The overall idea was to **allow** one application on **one computer** to talk to (send data packets) another application running on **different computer**.

► TCP/IP Model :

- The TCP/IP model is a widely used networking model that consists of **four layers:** Application, Transport, Internet, and Link(Network Access Layer).
- TCP/IP is a set of protocols developed to allow cooperating computers to share resources across a network.
- It serves as the **foundation** for the **modern internet** and is responsible for the **end-to-end** communication of **data packets** across networks.
- The TCP/IP model **combines** the OSI model's physical and data link layers into a single "Link Layer" due to its focus on **real-world networking**.
- Key protocols within the TCP/IP model include **TCP** (Transmission Control Protocol) and **IP** (Internet Protocol), which enable reliable communication and routing of data packets.
- It is a more **flexible and adaptable** model compared to OSI, making it well-suited for the dynamic and evolving nature of the internet.

► TCP/IP Model :



► TCP/IP Model :

Layers

- Application Layer
- Transport Layer
- Internet Layer
- Network Access Layer/Link layer

PDUs

- Application message
- TCP segment/UDP datagram
- IP packet
- Ethernet Frame

► TCP/IP Model :

- **Application Layer:**

- This layer is responsible for providing network services directly to **end-users** or applications. It includes protocols like **HTTP** (for web browsing), **SMTP** (for email), **FTP** (for file transfer), and more. Application layer protocols facilitate communication between **software applications** running on different devices.

- **Transport Layer:**

- The transport layer is responsible for **end-to-end** communication and **data flow control** between devices on different networks. It includes two main protocols: Transmission Control Protocol (**TCP**) and User Datagram Protocol (**UDP**). TCP provides reliable, connection-oriented communication, while UDP offers connectionless, unreliable communication.

- **Internet Layer/Network Layer:**

- This layer handles **logical addressing, routing, and forwarding of data packets** across different networks. The primary protocol at this layer is the Internet Protocol (**IP**), which enables the routing of packets from the source to the destination based on IP addresses.

- **Link Layer:/Network Access Layer**

- The link layer, sometimes called the **network interface or data link layer**, deals with the **physical connection** between devices on the same local network segment. It includes hardware-specific protocols and technologies, such as Ethernet, Wi-Fi, and others. This layer is responsible for **addressing and error detection** at the local network level.

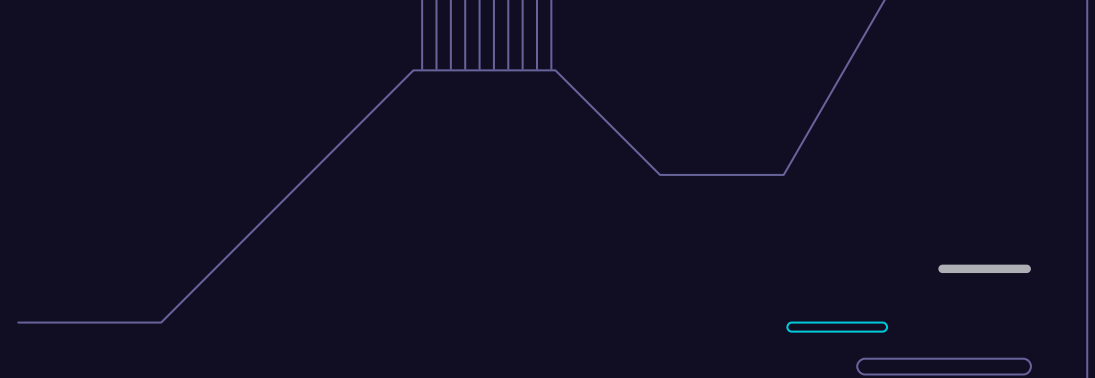
► Merits of TCP/IP Model:

- It operated independently.
- It is scalable.
- Client/server architecture.
- Supports a number of routing protocols.
- Can be used to establish a connection between two computers.

OSI vs TCP/IP Model :

Parameters	OSI Model	TCP/IP Model
Full Form	OSI stands for Open Systems Interconnection.	TCP/IP stands for Transmission Control Protocol/Internet Protocol.
Layers	It has 7 layers.	It has 4 layers.
Usage	It is low in usage. Generally in Research.	It is mostly used for network design.
Model	It is theoretical/Reference Model.	It is Practical/Implementation Model.
Delivery	Delivery of the package is guaranteed in OSI Model.	Delivery of the package is not guaranteed in TCP/IP Model.
Replacement	Replacement of tools and changes can easily be done in this model.	Replacing the tools is not easy as it is in OSI Model.
Reliability	It is less reliable than TCP/IP Model.	It is more reliable than OSI Model.
Developed By	ISO(International Organization for Standardization)	DoD(Department of Defense)
Model Concept	Based on three concept i.e Service, Interface and protocols.	It didn't explicitly distinguish between service ,interface and protocols.
Layer Separation	OSI model has separate Presentation layer and Session layer.	TCP/IP model does not have separate Presentation layer and Session layer.

▶ **Connectionless &
Connection-oriented
Services**



1.8

► Connectionless and Connection-Oriented Network Service:

- ❖ A "connection" often refers to a link or bond between two or more things/
Devices.
 - **Example:** electrical connections, internet connections, or the links between different parts of a machine.
- ❖ Before communication, the user in connection oriented service follows following sequence of operations:
 - Connection is established.(Virtual Connection)
 - Data/Information Sent.
 - Connection Release.

► Connection–Oriented Network Service:

Definition: A network service designed to **establish a reliable end-to-end connection** before data transmission.

Key Characteristics:

- **Orderly Packet Delivery:** Packets arrive in the order they were sent.
- **Handshake Method:** Establishes connection using a handshake process.
- **Reliability:** Known as a reliable network service due to its structured communication.

Communication Process:

1. Connection Setup:

- **SYN Packet:** Sender sends a request to the receiver.
- **SYN-ACK Packet:** Receiver responds to confirm readiness for communication.

2. Data Transfer:

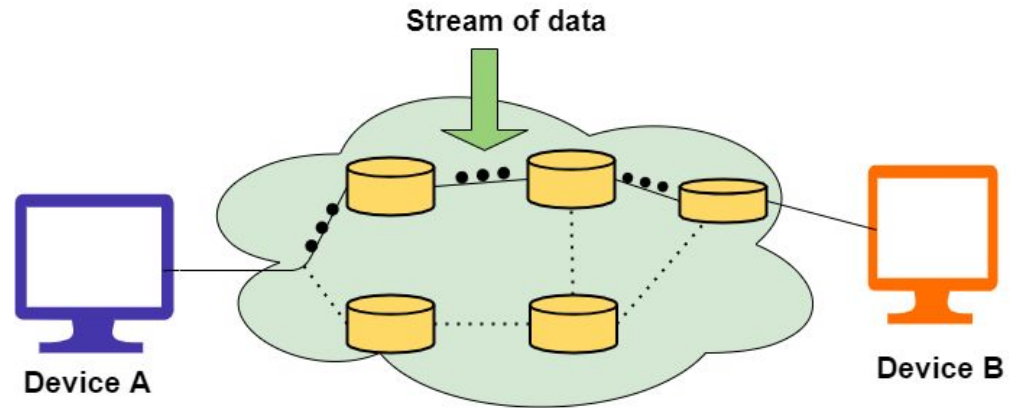
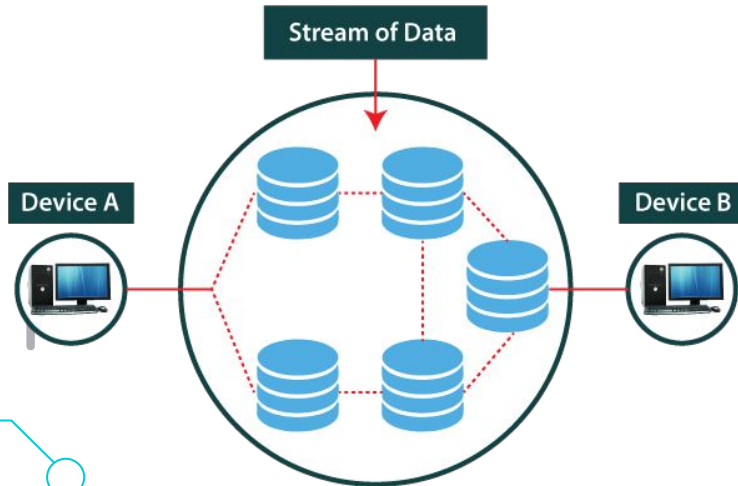
- Data packets are sent and received in the established order.
- Both sender and receiver can send and receive data packets.

3. Connection Termination:

- Sender terminates the connection after data exchange by sending a termination signal.

► Connection-oriented Network Service:

Connection-oriented Communication



Connectionless Network Service:

Definition: A network service that **does not establish** a dedicated end-to-end connection before transmitting data.

Key Characteristics:

- **No Setup Phase:** Data packets are sent independently without a prior handshake.
- **Packet Delivery:** Packets may arrive out of order, and some packets may be lost.
- **Best-Effort Delivery:** Offers no guarantee of delivery, order, or duplicate protection.
- **Lower Overhead:** Faster for small amounts of data as it avoids connection setup and termination overhead.

Communication Process

1. Data Transmission:

- Each packet contains the destination address and is routed independently.
- No acknowledgment of packet receipt is required.

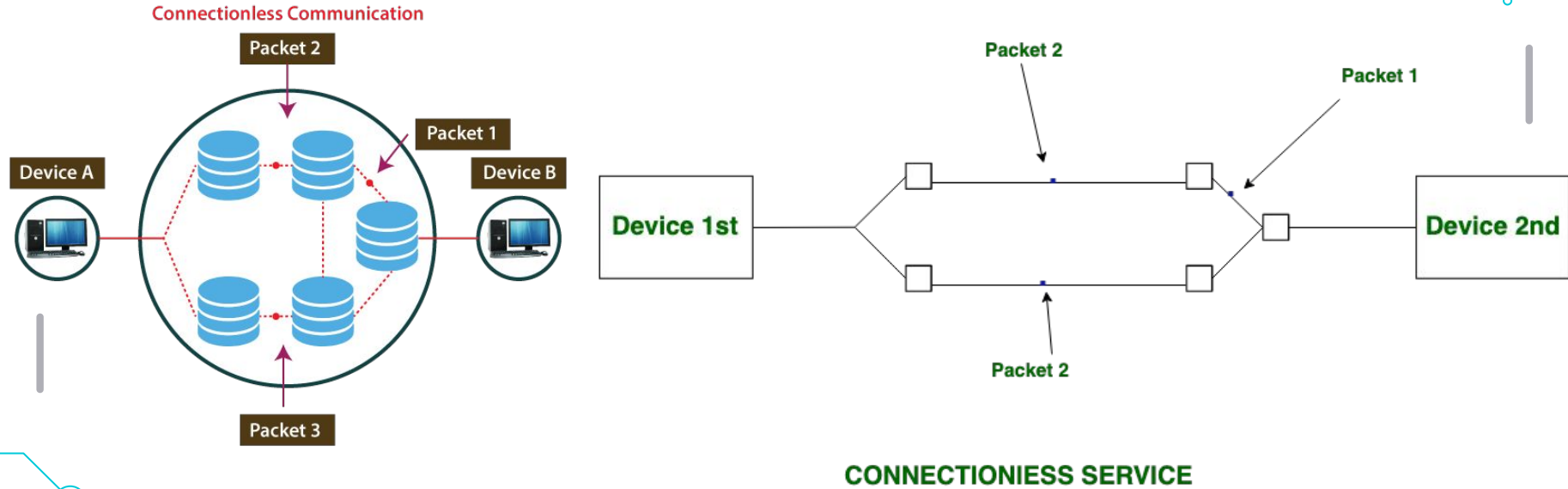
2. Packet Handling:

- Packets may take different paths to the destination.
- There is no guarantee that packets will arrive in the correct order or even arrive at all.

3. No Connection Termination:

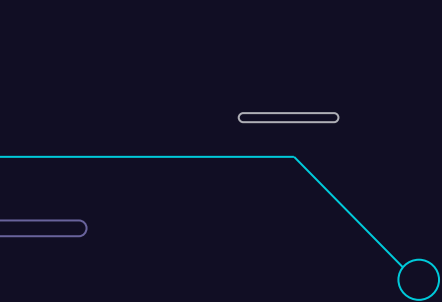
- Since there is no established connection, termination is not necessary.

► Connectionless Network Service:

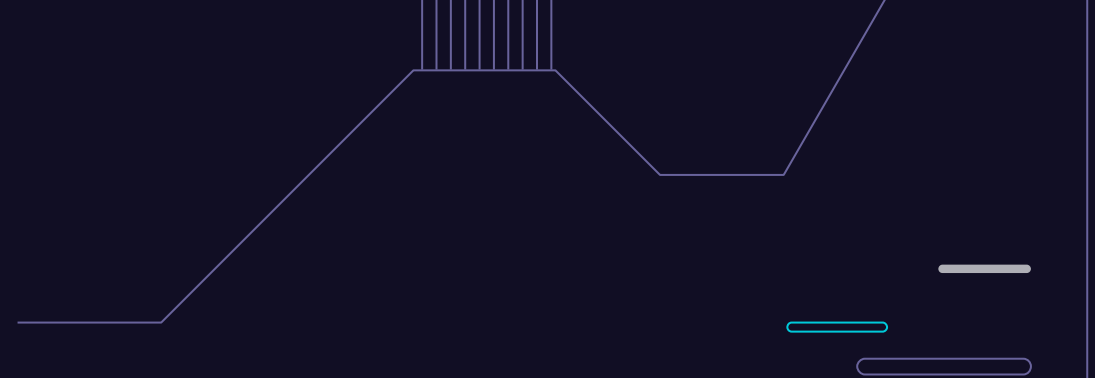


► Connectionless and Connection-oriented Network Service:

Parameter	Connection-Oriented	Connection-Less
Connection	Prior connection need to be established.	No prior connection is established.
Resource Allocation	Resource need to be allocated.	No prior allocation of resource is required.
Reliability	It ensures reliable transfer of data.	Less reliable.
Congestion	Congestion is not at all possible.	Congestion can occur likely.
Transfer mode	It can be implemented either using Circuit switching or VCs.	It can be implemented using Packet Switching.
Retransmission	It is possible to retransmit the lost data bits.	Not possible
Suitability	It is suitable for long and steady communication.	It is suitable for <u>bursty</u> transmissions.
Signalling	Connection is established through process of signalling.	There is no concept of signalling.
Packet travel	Sequential fashions	Random manner
Delay	More delay	NO delay due to absence of connection established phase.



Internet, ISPs, Backbone Network Overview



1.9

Internet:

The internet is a global network of interconnected computers and computer networks.

Internet offers:

- **Global Connectivity:** The internet connects millions of devices worldwide, facilitating communication and data exchange on a global scale.
- **Decentralized Structure:** It operates on a decentralized architecture, meaning there is no central control or single point of failure. Information is distributed across a multitude of servers and networks.
- **Protocols and Standards:** The Internet relies on standardized protocols (e.g., TCP/IP) and open standards to ensure compatibility and seamless communication between diverse hardware and software.
- **Diverse Services:** The Internet provides a diverse range of services, including web browsing, email, file sharing, social networking, and more. Users connect through Internet Service Providers (ISPs) using various technologies like broadband, wireless, or dial-up.

► ISPs (Internet Service Provider):

- ❖ An Internet Service Provider (ISP) is a company or organization that provides users with **access** to the Internet.
- ❖ In the context of Internet Service Providers (ISPs), the terms "**Tier 1**," "**Tier 2**," and "**Tier 3**" are often used to describe **different levels** of Internet connectivity and network infrastructure. These tiers are based on the **hierarchy and the scope** of the networks they operate.
- ❖ Internet service providers may be organized in various forms, such as **commercial**, **community-owned**, **non-profit**, or otherwise **privately owned**.
- ❖ Internet **services typically provided by ISPs include** Internet access, Internet transit, domain name registration, web hosting, Usenet service etc.
- ❖ Example: WorldLink, Vianet, NTC

► ISPs & Tiers:

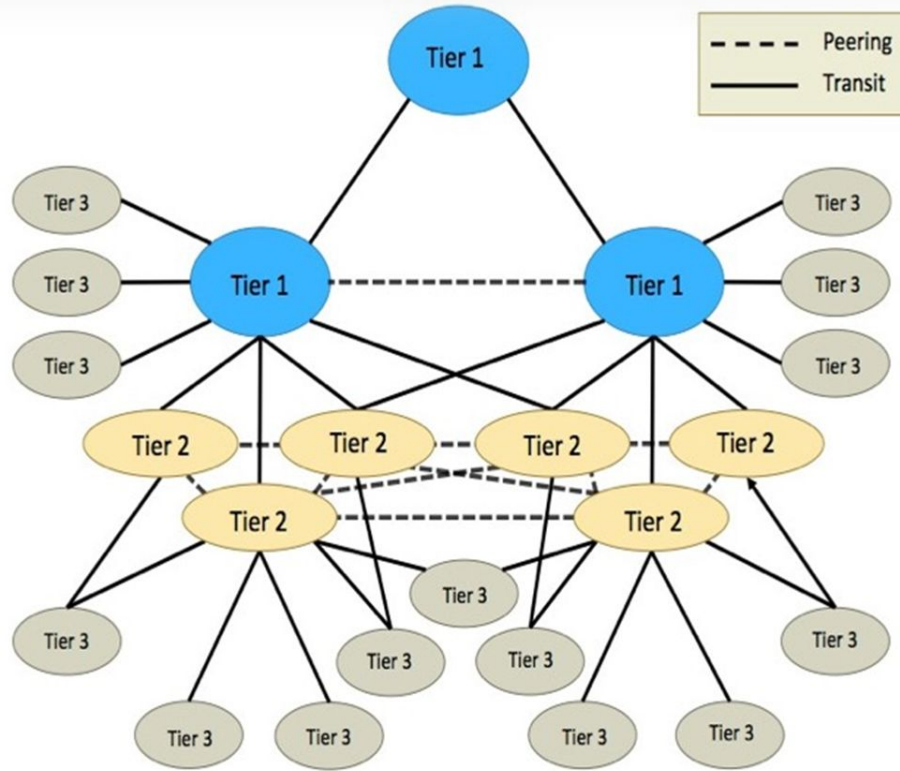


Fig: Tier Hierarchy of ISPs

► Tier-1:

- Tier 1 ISPs are considered the **top level** in the ISP hierarchy.
- They have a global network infrastructure that covers **multiple continents** and regions.
- Tier 1 ISPs are the **backbone** of the internet, providing high-speed connectivity on a global scale.
- They handle large volumes of internet traffic and play a crucial role in the stability and performance of the internet.
- Examples of Tier 1 ISPs include **AT&T, Verizon**.

► Tier-2:

- Tier 2 ISPs operate at a **regional or national** level.
- They have a network infrastructure that covers a **specific geographic area or country**.
- Tier 2 ISPs act as **regional providers**, connecting end-users, businesses, and other networks to the **broader internet**.
- They may serve as **intermediaries** between Tier 1 ISPs and smaller, local ISPs.
- Examples of Tier-2 ISPs include Cogent Communications , Zayo Group etc.

► Tier-3:

- Tier 3 ISPs are typically **smaller** and operate at a **local** or **community** level.
- Tier 3 ISPs may connect to **larger ISPs** or **Tier 2** ISPs to **access** the **global** internet.
- Tier 3 ISPs are often the **last mile** providers, delivering internet connectivity **directly** to **end-users' homes** or businesses.
- They play a crucial role in providing local access and connectivity to smaller communities.
- Examples: Airtel, Worldlink etc

► Types of ISPs:

◆ Hosting ISPs:

- Hosting Internet Service Providers specialize in providing **web hosting services**, offering **server space** and **infrastructure** for individuals and businesses to host websites and applications on the internet.

◆ Mobile box ISPs:

- Mobile Box Internet Service Providers typically refer to **telecommunications companies** that provide **internet** services through **mobile networks**. Users connect to the internet using mobile devices, such as smartphones and tablets.

◆ Transit ISPs.

- Transit Internet Service Providers **operate at the core** of the internet, providing **high-level connectivity** between different networks. They facilitate the transit of data between networks, often through extensive **peering agreements with other ISPs**.

► Types of ISPs:

◆ Virtual ISPs:

- Virtual Internet Service Providers are companies that **lease network infrastructure from other ISPs** rather than owning **their own** physical network. They offer internet services to **end-users without owning** the underlying network infrastructure.

◆ Free ISPs:

- Free Internet Service Providers offer internet access to users at **no direct cost**. These services are often **ad-supported**, generating **revenue** through **advertising** to offset the expenses of providing free internet access.

◆ Wireless ISPs.

- Wireless Internet Service Providers deliver internet connectivity to users using **wireless technologies**, such as **Wi-Fi or fixed wireless** access. These services are often **deployed in areas** where **traditional wired** connections may be **challenging or unavailable**.

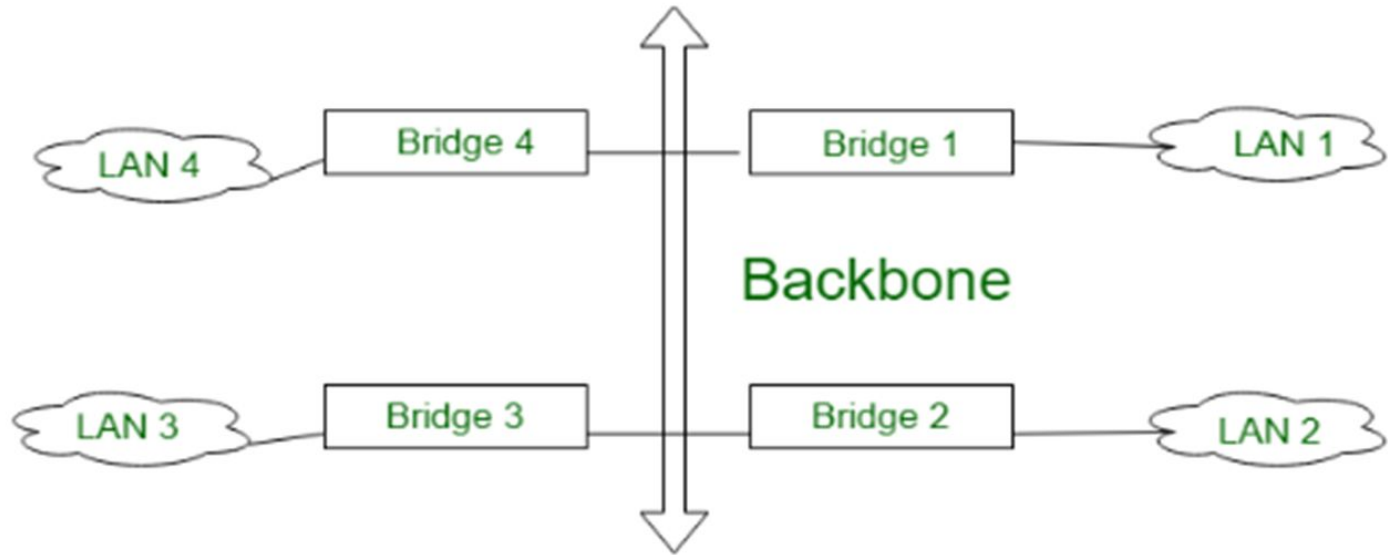
► Backbone Network:

- ❖ A Backbone Network, often referred to as a **core network**, is a high-capacity infrastructure that serves as the **primary pathway** for data transmission between **different network segments** and major locations within a network.
- ❖ It forms the **central and essential part** of a telecommunications or computer network, providing a robust and efficient means of **interconnecting various subnetworks**.
- ❖ The Backbone LAN serves as a **high-capacity interconnection** between the individual LANs in different buildings or departments. It is designed to handle the **aggregated data traffic** from **multiple sources** and facilitate **efficient communication between the various LANs**.

► Types of Backbone Network:

- ❖ The backbone **itself** is a LAN, it uses a **LAN protocol** such as Ethernet, Hence **each connection** in the backbone is itself another LAN.
- ❖ The commonly used architectures are:
 - Bus backbone
 - Star backbone
 - Interconnecting a remote LAN
- ❖ **Bus backbone**
 - In Bus backbone the **topology** used for the backbone is **bus topology**.
 - Bus backbones are normally used as a **distribution backbone** to connect different buildings in an organization.

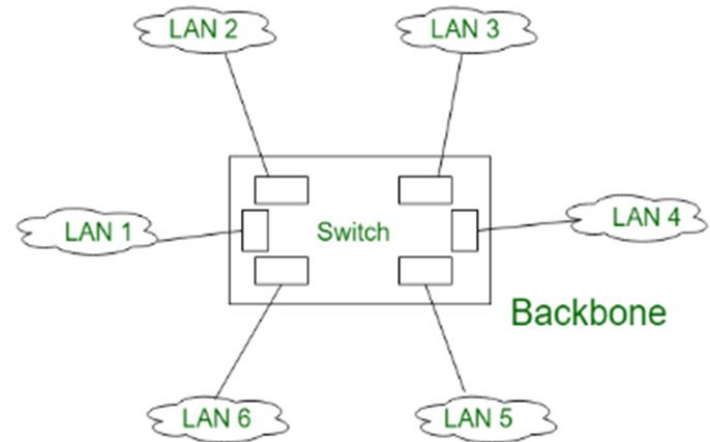
► Bus Backbone:



Structure of a Bus backbone

► Star Backbone:

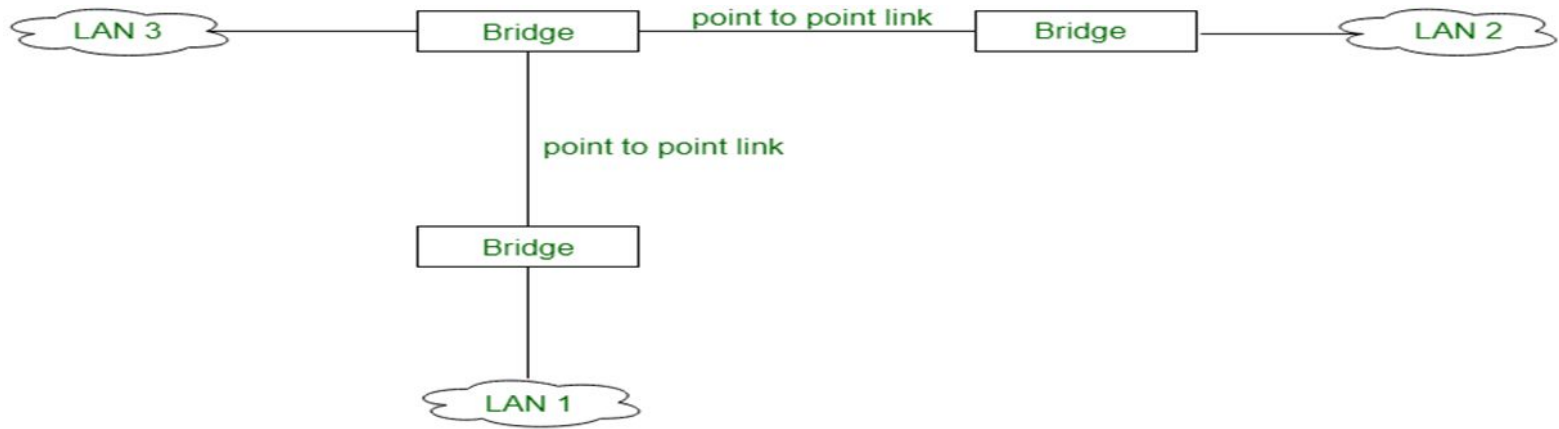
- The topology of this backbone is **star topology**.
- In a star backbone, sometimes called **switched** backbone, the topology of the backbone is a star. In this configuration, the backbone is just **one switch that connects the LANs**.
- In this configuration, the switch **does the job of** the backbone and at the same time **connects** the LANs.



Star Backbone

► Interconnecting a Remote LAN

- In this type of backbone network, connections are established through **remote bridges**, serving as connecting devices that link LANs via point-to-point network links.



Connecting remote LANs to each other

Growth of the Internet:

The **increase** in the number of networks and connected computers has been **substantial**, moving from tens to hundreds of thousands of networks and from hundreds to hundreds of millions of connected devices.

New Protocols:

- Continuous development and implementation of new protocols are crucial.
- Deprecated protocols need to be phased out.
- **Example:** IPv6, which offers many improvements over IPv4, has been approved as a standard but is not yet fully implemented.

New Technology:

- Advancements in technology will enhance **network capacity and bandwidth**.
- These developments are aimed at accommodating more users and providing faster, more reliable Internet access.

Increasing Use of Multimedia:

- The Internet is increasingly used for multimedia purposes, such as audio and video.
- This shift from merely sharing data to supporting rich multimedia content drives further growth and innovation.

► **Presentation Session:**

❖ *Presentation Date: 24th May, 2026*

❖ *Duration: 20 minutes (maximum) / group*

❖ **Topic-1: OSI Model & Comparison with TCP/IP Model**

➤ **Team Members:**

- Suman Adhikari
- Karun Gyawali

❖ **Topic-2: Network Topologies**

➤ **Team Members:**

- Sharas Ghatani
- Anish Shiwakoti

► PYQs:

- ❖ Explain about connection-oriented network services. 2081
- ❖ Difference between web server and proxy server. 2081
- ❖ Explain different types of network topologies. 2080
- ❖ Explain different types of network types. 2080
- ❖ Short Notes: 2079
 - Protocol and standards
 - Backbone Network
- ❖ Define Protocol. Why do we need standards? 2079
- ❖ Why do we need network topology? Explain star topology along with its merits and demerits. 2079
- ❖ What is protocol? Explain each layer of OSI model in detail. 2079
- ❖ Explain each layer of TCP/IP model in detail. Compare it with OSI model. 2078
- ❖ What are the two network architectures in practice explain with example?
- ❖ Classify network in the basis of size.
- ❖ Explain client/server network. How is it different from peer to peer network? 2081
- ❖ Explain LAN with example. How is it different from PAN?

► Assignment-1:

Due Date: 29th May 2026

- ❖ Explain about connection-oriented network services. 2081
- ❖ Difference between web server and proxy server. 2081
- ❖ Explain different types of network topologies. 2080
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► **THANKS!**

? **Do you have any questions?**

✉ **shishir.ghimire@newsummit.edu.np**